

Unified Combinatorial Coefficient Calculus

**Stirling, Bell, Eulerian, Bernoulli, Euler, and Genocchi Families;
Faà di Bruno Composition, Lagrange–Good Inversion,
Pochhammer–Polygamma Calculus, Barnes’ G -Function, and
Asymptotics**

A merged, deduplicated, and proof-complete synthesis

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Abstract

This monograph merges five overlapping draft articles into a single coherent coefficient calculus. Repetition has been removed theorem by theorem: each central definition and identity appears once, with the strongest available hypotheses, the clearest normalization, and a detailed proof. Material that was genuinely complementary—alternative proofs, algorithms, applications, asymptotic methods, and special-function connections—has been integrated where it belongs mathematically rather than appended as disconnected source notes.

The common core comprises signed and unsigned Stirling numbers of the first kind, Stirling numbers of the second kind, Lah numbers, Bell numbers and ordinary and exponential Bell polynomials, Eulerian numbers of types A and B , second-order Eulerian numbers, Faà di Bruno’s formula, and one- and several-variable Lagrange inversion. The unified treatment develops these objects from formal power series, finite differences, labelled combinatorial classes, triangular changes of basis, incidence algebras, exponential Riordan arrays, and formal residues.

The article substantially extends the drafts using the supplied [Wikipedia/](#) and [MathWorld/](#) collections and a targeted review of authoritative and primary online sources. New proof-complete chapters cover partition-lattice Möbius inversion; moment–cumulant relations; Sheffer and binomial-type sequences; Bernoulli polynomials of both kinds; Euler and Genocchi polynomials; complementary Bell numbers; Darboux’s weighted Taylor formula and Darboux coefficient transfer; Euler–Maclaurin summation with an explicit periodic-Bernoulli remainder; Pochhammer parameter derivatives expressed through Bell polynomials; digamma and polygamma multiplication and reflection; Barnes’ G -function and its Euler–Maclaurin asymptotics; multivariate Lagrange–Good inversion; Narayana refinements; Kepler’s equation; and the universal square-root singularity of a locally inverted map at a simple critical point.

Formal and analytic assertions are kept distinct. Convergence domains, branch choices, boundary conventions, determinant factors, and remainder terms are stated explicitly. A final merge ledger identifies the role of every draft and every supplemental topic, records corrections and deduplication decisions, and supplies a reproducible dependency map for the resulting article.

How to read this work

The chapters are arranged by mathematical dependence, not by the order of the five source drafts. The first major sequence builds the shared combinatorial core; the later parts develop the incidence-algebra, Bernoulli–Euler, special-function, and multivariate-inversion extensions. The two master ideas are:

1. a coefficient is often easiest to compute after changing to the basis in which the relevant operator is diagonal; and
2. an exponential generating function remembers the decomposition of a labelled object into components.

Stirling numbers implement the first idea. Bell polynomials implement the second. Eulerian numbers mediate between powers and shifted binomial coefficients. Faà di Bruno’s formula is the differential form of labelled composition, while Lagrange inversion is the coefficient form of implicit substitution.

A result stated for formal power series requires no convergence. When an analytic version is asserted, its domain hypotheses are written explicitly. Statements in the archive that report historical computations—for example, lists of Bell primes or verified exact modular periods—are distinguished from theorems and are accompanied by a certificate procedure rather than misrepresented as symbolic identities.

Editorial normalization

Three systematic repairs are used throughout.

1. Bell numbers are denoted B_n , complete exponential Bell polynomials by $\mathcal{B}_n$, partial Bell polynomials by $\mathcal{B}_{n,k}$, and Bernoulli numbers by β_n .
2. Type- B Eulerian numbers are indexed by descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift in one formula in the archive.
3. The archived Moser–Wyman display contains unnamed quantities B, P_i, Q_i . We give the exact saddle-point scheme that defines every correction coefficient recursively and state the leading expansion in closed form; no undefined placeholder is retained.

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Notation

For $n \in \mathbb{N}$, write

$$x^{\overline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\underline{n}} = x(x+1) \cdots (x+n-1),$$

with both expressions equal to 1 when $n = 0$. Coefficient extraction is denoted by $[z^n]F(z)$, and $\text{Res}_z F(z) dz$ means the coefficient of z^{-1} in a formal Laurent series. Our principal arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle,$$

where $s(n, k)$ is the signed first-kind Stirling number, $\begin{bmatrix} n \\ k \end{bmatrix}$ its unsigned version, $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ the second-kind Stirling number, and $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ the ordinary Eulerian number. Bell numbers are B_n ; exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$.

Unless stated otherwise, sums over an impossible index range are zero, and $0^0 = 1$ in coefficient formulae. Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). *For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,*

$$[z^n]F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n]F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}]A(z)$. Then

$$[z^n]F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.2 (Formal change of variables for residues). *Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,*

$$\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + O(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$\hat{A}(z)\hat{B}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta x^{\underline{k}} = kx^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $x + 1^{\underline{k}} - x^{\underline{k}} = kx^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.3 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} x^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k x^{\underline{k}}$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n(x_1, \dots, x_n)$.

Theorem 1.4 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Corollary 1.5 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$x^n = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$x^{\bar{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $x^{\bar{n}} = (-1)^n - x^n$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $x^{\bar{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n=3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x=1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). For $n, k \geq 0$,

$$\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). For all $n, k \geq 0$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

The simple diagonals follow at once:

$$\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1, \quad \left\{ \begin{matrix} n \\ 1 \end{matrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{matrix} n \\ n-1 \end{matrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.6 (Alternative second-kind recurrences). For $1 \leq k < n$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{matrix} n \\ k+j-1 \end{matrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{matrix} n-j+1 \\ k \end{matrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{matrix} n \\ r \end{matrix} \right\} k^r$. Since $k^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\binom{-k}{j} = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.7 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = y^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{y^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x) \log(1-x)) C'_k(x).$$

The left side has generating function $xC'_k - kC_k$. Their equality reduces to

$$-(1-x) \log(1-x) C'_k(x) = kC_k(x),$$

which follows immediately from the displayed formula for C_k . □

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6
0	1						
1	0	1					
2	0	1	1				
3	0	2	3	1			
4	0	6	11	6	1		
5	0	24	50	35	10	1	
6	0	120	274	225	85	15	1

$n \backslash k$	0	1	2	3	4	5	6
0	1						
1	0	1					
2	0	1	1				
3	0	1	3	1			
4	0	1	7	6	1		
5	0	1	15	25	10	1	
6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n \}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $x^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Corollary 3.2 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} x - 1^{\underline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} n^{\underline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $x^{\underline{k}} = x x - 1^{\underline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Corollary 3.3 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\{n_k\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand x^n into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{N+1^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2}N + 1^2 + \frac{7}{3}N + 1^3 + \frac{6}{4}N + 1^4 + \frac{1}{5}N + 1^5,$$

which expands to the familiar quartic power-sum polynomial.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.4 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}X^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{r^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} X^{\underline{k}}$. $\square$

Theorem 3.5 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n .

Proof. The random variable Y_m^k counts ordered k -tuples of distinct fixed points. There are m^k candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}Y_m^k = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}Y_m^k = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. $\square$

3.4 Lah numbers

Definition 3.6. For $1 \leq k \leq n$, the unsigned Lah number is

$$L(n, k) = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $L(0, 0) = 1$ and zero otherwise outside the triangle.

Combinatorially, $L(n, k)$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.7 (Conversion between rising and falling factorials).

$$x^{\bar{n}} = \sum_{k=0}^n L(n, k) x^k, \quad (3.16)$$

$$x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\bar{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} L(n, j) L(j, k) = \delta_{nk}. \quad (3.18)$$

Moreover,

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} L(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k L(n, k) x^k \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} x^{\bar{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq L(n, k) \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.2 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} 1/x^{k+1}}. \quad (4.6)$$

Consequently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\begin{Bmatrix} n \\ r \end{Bmatrix}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ by $k!$.

4.3 The Stirling transform

Theorem 4.3 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$\widehat{G}(z) = \widehat{F}(e^z - 1), \quad \widehat{F}(z) = \widehat{G}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.3. For the generating function,

$$\widehat{G}(z) = \sum_k f_k \sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = \widehat{F}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $m^{\underline{k}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.4 (Stirling normal-ordering identities). *For $n \geq 0$,*

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = x D^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^{\underline{n}} = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} m^{\underline{k}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $m^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, 2 + 2 for the second diagonal and 4, 3 + 2, 2 + 2 + 2 for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m[x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ n \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ j \end{matrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx} x^{\bar{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X+c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.2 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j-k$ and lower index $j-n$:

$$\begin{Bmatrix} j-k \\ j-n \end{Bmatrix} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-m-n+k)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.3 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{\prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p=0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p=0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1+A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$x^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $x^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $x^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

6.2 Parity of second-kind numbers

Theorem 6.2 (Parity criterion). *Let*

$$z = n - \left\lceil \frac{k+1}{2} \right\rceil, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv \binom{z}{w} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\left\{ \begin{matrix} 2n \\ n \end{matrix} \right\}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\binom{n-k+\lceil k/2 \rceil-1}{\lceil k/2 \rceil-1} = \binom{z}{w}$. Lucas's theorem modulo 2 says $\binom{z}{w}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \left\{ \begin{matrix} -k \\ -n \end{matrix} \right\}, \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left[\begin{matrix} -k \\ -n \end{matrix} \right]. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\left[\begin{matrix} -n \\ k \end{matrix} \right] = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\left[\begin{matrix} -5 \\ k \end{matrix} \right] = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x)\cdots(n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$B_{-1} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$. $\square$

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = [t^k] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $O(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$T_{n+1}(x) = x(T_n(x) + T'_n(x)), \quad T_0(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of T_n are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $T_1(x) = x$. Suppose $f = T_n$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + O(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + O(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

Let $v = n/k > 1$ and let $G \in (0, 1)$ be the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Equivalently, if $r = v - G$, then $r/(1 - e^{-r}) = v$. Saddle-point extraction from $(e^z - 1)^k/k!$ gives

$$\left\{ \frac{n}{k} \right\} \sim \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k}, \quad (7.13)$$

uniformly away from the two extreme edges; refined versions have relative error $O(1/n)$, and the numerical constant $0.066/n$ quoted in the archive belongs to one particular explicit error theorem.

Derivation. By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi i k!} \int \exp(k \log(e^z - 1) - n \log z) \frac{dz}{z}.$$

The positive saddle r solves $ke^r/(e^r - 1) = n/r$, or $r/(1 - e^{-r}) = v$. Put $G = v - r$ to obtain Equation (7.12). Expanding the phase to second order at r , evaluating the Gaussian integral, and simplifying with Stirling's formula for $n!/k!$ gives exactly Equation (7.13). On a fixed angular complement of the saddle the real part of the phase drops quadratically; Taylor's theorem with an explicit third- and fourth-derivative bound gives a relative $O(1/n)$ remainder. Thus the formula is a theorem of the saddle method, not a heuristic replacement of the generating function. $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.4 (Stirling-zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n v^{\bar{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i / (e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/v^{\bar{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{v^{\bar{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m} / (e^u - 1) du = \Gamma(z+1-m) \zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\zeta(s, v) = \frac{k!}{s - k} \sum_{n \geq 0} \frac{\begin{bmatrix} n+k \\ n \end{bmatrix}}{(n+k)!} \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.5 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $O(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^w$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z+w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.6 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.22)$$

Proof. Put $L(u) = -\log(1-u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell+1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right], \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \left[\begin{matrix} n \\ r \end{matrix} \right] u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1-u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.5 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell+1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.7 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m = & \frac{1}{2}\delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ & \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^{\infty} \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^{\infty} \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1+w)}{1+w} = m! \sum_{r \geq m} s(r+1, m+1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k+2, m+1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k+1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k+1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k+1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.5 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno’s formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n are defined by

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} G_n z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1+z)$ gives

$$n!G_n = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell+1}. \quad (7.30)$$

Indeed, $z/\log(1+z) = \int_0^1 (1+z)^t dt$ and comparison with $(1+z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n \mu^k s(n, n-k+1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}D_t^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand D_t^n by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N} \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := [x^{k-1}](x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.2 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n + 1$. If deleting $n + 1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n + 1$ has exactly r elements: choose its $r - 1$ companions and partition the remaining $n - r + 1$ labels into $k - 1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j / j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.3 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n - d + 1 \\ k - d + 1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i - j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) q^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d - 1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d - 1$ vertices, which form a clique, and hence has $q - d + 1$ choices. Thus

$$\chi_{G_{n,d}}(q) = q^{d-1} (q - d + 1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned} \chi_{G_{n,d}}(q) &= q^{d-1} \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{-d+1+r} \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r+d-1}. \end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). $\square$

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0 = B_1 = 1, \quad (B_n)_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.4. *If $N = p_1 \cdots p_n$ is squarefree, then B_n is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.5. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.6. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.7 (Return-to-order shuffles). *Exactly B_n of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.8. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.9. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{ccccccc} 1 & & & & & & \\ 1 & 2 & & & & & \\ 2 & 3 & 5 & & & & \\ 5 & 7 & 10 & 15 & & & \\ 15 & 20 & 27 & 37 & 52 & & \end{array}$$

Theorem 9.10. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n$ and the right edge is $a_{n,n} = B_{n+1}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). For $n \geq 0$,

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

10.2 Spivey's identity and binomial inversions

Theorem 10.3 (Spivey's identity). For $m, n \geq 0$,

$$B_{m+n} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.4 (Bell inversion identities). For $n, k \geq 0$,

$$B_n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.7 below gives $\mathbb{E}X^r = B_r$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.5 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}. \quad (10.9)$$

Proof. By Dobinski's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} m^i = m^{\underline{k}}$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{m^{\underline{k}} [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

10.3 Exponential generating function and differential equation

Theorem 10.6. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

10.4 Dobiński's formula and Poisson moments

Theorem 10.7 (Dobiński). *For every $n \geq 0$,*

$$B_n = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z-1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a Poisson(1) variable. $\square$

Corollary 10.8 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. $\square$

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n to the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n(1) = B_n.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}(x) \equiv T_{n+1}(x) + x^p T_n(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p} \equiv B_n + B_{n+1} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Theorem 11.2 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m} \equiv m B_n + B_{n+1} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}(x) \equiv T_{n+1}(x) + T_n(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1} + T_n \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

11.2 A universal period bound modulo a prime

Theorem 11.3. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.4. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.5. *For $n \geq 1$,*

$$1 \leq \frac{B_{n-1}B_{n+1}}{B_n^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n) is log-convex, whereas $(B_n/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1}\mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{t-1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1} z_{j+1}) = j/(j-1)$. Hence $(B_n/n!)^2 \geq (B_{n-1}/(n-1)!)(B_{n+1}/(n+1)!)$, which is the right inequality in (11.6). $\square$

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.6. *For $t > 0$,*

$$B_n \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$B_n^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.7 (Uniform numerical bounds). *For every $n \geq 1$,*

$$B_n < \left(\frac{0.792 n}{\log(n+1)} \right)^n. \quad (11.9)$$

Define, for $x > 1$,

$$d(x) = \log \log(x+1) - \log \log x + \frac{1 + e^{-1}}{\log x}. \quad (11.10)$$

For every $\varepsilon > 0$ there is $n_0(\varepsilon) = \max\{e^4, d^{-1}(\varepsilon)\}$ such that, for $n > n_0(\varepsilon)$,

$$B_n < \left(\frac{e^{-0.6+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.11)$$

Proof. Set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

Direct differentiation, after multiplying by the positive denominator $2x \log x \log(x+1) \log(x/\log(x+1))^2$, shows that $R'(x) < 0$ for $x \geq 39$; substitution gives $R(39) < 0.792$. Thus (11.8) proves (11.9) for $n \geq 39$. The remaining $1 \leq n \leq 38$ are obtained successively from (10.1); substituting them gives strict inequalities. This is a finite exact integer check.

A second differentiation gives, for $x \geq e^4$,

$$\log R(x) \leq -0.6 + d(x), \quad d'(x) < 0, \quad \lim_{x \rightarrow \infty} d(x) = 0.$$

(The numerator after clearing positive denominators is a sum of $-(\log x - 4)^2$, $-(x - 1 - \log x)$, and negative multiples of these two standard nonnegative quantities.) Hence $d(n) < \varepsilon$ when $n > d^{-1}(\varepsilon)$, and (11.11) follows from (11.8). $\square$

11.5 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1+r)$.

Theorem 11.8 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \quad (11.12)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1\right). \quad (11.13)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n}{n!} = \frac{e^{e^r} - 1}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du/\sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.12). Stirling's formula and $e^r = n/r$ simplify it to (11.13). $\square$

The archived refined expansion used undefined symbols. The following construction makes every coefficient unambiguous. Let $T_j(r) = \sum_{q=0}^j \{j\}_q r^q$ be the Touchard polynomial and let $Z \sim N(0, 1)$. For an integer $h = O(r)$ define

$$\mathcal{R}_{r,h}(Z) = \exp\left(-\frac{ihZ}{\sqrt{A}} + \sum_{j \geq 3} \frac{i^j e^r T_j(r)}{j! A^{j/2}} Z^j\right). \quad (11.14)$$

Expand this formal exponential and take Gaussian moments termwise; retaining all monomials of total order at most e^{-Mr} defines $C_0(r, h), \dots, C_M(r, h)$, with $C_0 = 1$.

Theorem 11.9 (Uniform Moser–Wyman scheme). *For every fixed M and $|h| \leq Cr$,*

$$B_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi A}} \left(\sum_{q=0}^{M-1} C_q(r, h) + O_C(e^{-Mr} P_M(r)) \right), \quad (11.15)$$

where P_M is an explicitly computable polynomial. The first correction is

$$C_1(r, h) = -\frac{h^2}{2A} - \frac{he^r T_3(r)}{2A^2} + \frac{e^r T_4(r)}{8A^2} - \frac{5e^{2r} T_3(r)^2}{24A^3}. \quad (11.16)$$

In particular the coefficient of e^{-qr} is a polynomial in h of degree at most $2q$, with rational functions of r as coefficients.

Proof. Use the same circle as in the preceding proof but replace n by $n+h$. After $\theta = Z/\sqrt{A}$, the ratio of the integrand to its Gaussian quadratic part is exactly the formal series (11.14). Watson's lemma with a Gaussian majorant permits termwise integration on the central arc; the complementary arcs have the same exponential bound uniformly for $|h| \leq Cr$. Odd Gaussian moments vanish, so only integral powers of e^{-r} remain. Since $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, the terms from the square of the h -term, its product with the cubic term, the quartic term, and the square of the cubic term give (11.16). The same finite bookkeeping defines every subsequent C_q and bounds the omitted Taylor remainder. $\square$

Corollary 11.10 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n}{n} &= \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ &\quad + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.17)$$

Proof. Take logarithms in (11.13). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + r \log \log n \log n + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $O(\log n/n)$. $\square$

11.6 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0 = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\hat{\mathcal{B}}_{n,k}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\hat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $\hat{X}(t) = \sum_{j \geq 1} x_j t^j$,

$$\hat{X}(t)^k = \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{u\hat{X}(t)} = \sum_{n \geq k \geq 0} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1} &= x_1, \\
\mathcal{B}_{2,1} &= x_2, & \mathcal{B}_{2,2} &= x_1^2, \\
\mathcal{B}_{3,1} &= x_3, & \mathcal{B}_{3,2} &= 3x_1x_2, & \mathcal{B}_{3,3} &= x_1^3, \\
\mathcal{B}_{4,1} &= x_4, & \mathcal{B}_{4,2} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3} &= 6x_1^2x_2, & \mathcal{B}_{4,4} &= x_1^4, \\
\mathcal{B}_{5,1} &= x_5, & \mathcal{B}_{5,2} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4} &= 10x_1^3x_2, & \mathcal{B}_{5,5} &= x_1^5, \\
\mathcal{B}_{6,1} &= x_6, & \mathcal{B}_{6,2} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5} &= 15x_1^4x_2, & \mathcal{B}_{6,6} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0 &= 1, \\
\mathcal{B}_1 &= x_1, \\
\mathcal{B}_2 &= x_1^2 + x_2, \\
\mathcal{B}_3 &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4 &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5 &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6 &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7 &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}, \quad (13.2)$$

with $\mathcal{B}_{0,0} = 1$, $\mathcal{B}_{n,0} = 0$ for $n > 0$, and $\mathcal{B}_{0,k} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}(x) \mathcal{B}_{n-i,k_2}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B(|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1} = x_n, \quad \mathcal{B}_{n,2} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n} = x_1^n, \quad \mathcal{B}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (1.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.5. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t))^2 F(t).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}(1, 1, \dots, 1) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (14.3)$$

$$\mathcal{B}_n(1, 1, \dots, 1) = \mathcal{B}_n, \quad (14.4)$$

$$\mathcal{B}_{n,k}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = L(n, k), \quad (14.5)$$

$$T_n(x) = \mathcal{B}_n(x, x, \dots, x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $T_n(x)$. $\square$

14.2 Homogeneity and a useful evaluation

Theorem 14.2 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}(1, 2, 3, \dots, n-k+1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k}k^{n-k}$. $\square$

14.3 Addition and shifted-zero identities

Theorem 14.3 (Complete Bell addition formula).

$$\mathcal{B}_n(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}(x) \mathcal{B}_i(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.4 (Insertion of q leading zeros). For $q \geq 0$,

$$\begin{aligned} \mathcal{B}_{n,k} \left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots \right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

14.4 Exponential–logarithmic inversion

Theorem 14.5 (Bell transform and its inverse). Suppose

$$y_n = \mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n/n!$. $\square$

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno's formula in Equation (19.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n = (h_{ij})$ and $K_n = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1} x_2 & \binom{n-1}{2} x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1} x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.6 (Bell determinants).

$$\mathcal{B}_n(x_1, \dots, x_n) = \det H_n = \det K_n. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n = D(JH_n^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.7. *If $x^{k\Diamond}$ denotes the k -fold diamond power, then*

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k\Diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k\Diamond}$ is $X(t)^k$. Compare with (1.10). □

For example,

$$x\Diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3\Diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n \sum_{k=1}^n \frac{1}{k} \widehat{\mathcal{B}}_{n,k}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k = -(k-1)! \operatorname{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n(s_1, s_2, \dots, s_n). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \operatorname{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.5. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n , $\mathcal{B}_n$, $\beta_n(x)$, and C_n are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$A_0(t) = 1, \quad A_n(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$A_n(t) = (1 + (n-1)t)A_{n-1}(t) + t(1-t)A'_{n-1}(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1-\pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad A_n(t) = t^{n-1}A_n(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} A_n(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.4. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Theorem 16.5 (A second polynomial recurrence). For $n > 1$,

$$A_n(t) = \sum_{k=0}^{n-1} \binom{n}{k} A_k(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

16.3 Worpitzky's identity and sums of powers

Theorem 16.6 (Worpitzky). For every indeterminate x and $n \geq 1$,

$$x^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Corollary 16.7 (Power sums). For $m, n \geq 1$,

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z A_n(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \right\rangle_k z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.8.

$$A_n(t) = \sum_{k=0}^n \left\{ n \right\}_k t^k (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\{ n \right\}_k m^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} m^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $A_n(t) = \sum_k \left\{ n \right\}_k k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.9. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle n \right\rangle_k f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0, 1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle n \right\rangle_k / n!$.

Proof. The volume of $\{x \in [0, 1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle n \right\rangle_k$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.10. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n}{k}} = (n+1) \beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n A_n \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $A_n(t)$, so its h^* -vector is the n th Eulerian row.*

The regions $\Delta_{n,k}$ in Theorem 16.9 are half-open hypersimplices. Their normalized volumes are $n! \operatorname{vol} \Delta_{n,k} = \langle \begin{smallmatrix} n \\ k \end{smallmatrix} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ordered partitions into k blocks.

Theorem 17.2. *The h -polynomial of the simple polytope dual to the type- A permutohedron is $A_n(t)$.*

Proof. For a simple $(n-1)$ -polytope the h -polynomial is the transform $h(t) = \sum_F (t-1)^{\dim F}$ over faces of the dual simplicial complex. Grouping braid-fan faces by their k ordered blocks gives $\sum_k k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} (t-1)^{n-k}$, which is $A_n(t)$ by (16.12). $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_B(n, k)$ count signed permutations with k such descents, and set $M_n(t) = \sum_{k=0}^n E_B(n, k)t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_B(n, k) = (2k + 1)E_B(n - 1, k) + (2n - 2k + 1)E_B(n - 1, k - 1), \quad (18.1)$$

$$\sum_{m \geq 0} (2m + 1)^n t^m = \frac{M_n(t)}{(1 - t)^{n+1}}, \quad (18.2)$$

$$E_B(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j) + 1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n - 1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k + 1$ preserving slots for an object with k descents and $2n - 2k + 1$ creating slots for one with $k - 1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m + 1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m + 1)^n = \sum_{k=0}^n E_B(n, k) \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1 - t)^{n+1}$ to obtain (18.2). Multiplication by $(1 - t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

As in type A , these coefficients form the h -vector of the simple polytope dual to the type- B permutohedron; the proof repeats the braid-fan argument with signed ordered partitions.

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$P_n(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad P_0(t) = 1.$$

Theorem 18.2 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n - 1 \rangle\rangle_{k-1} + (k + 1) \langle\langle n - 1 \rangle\rangle_k, \quad (18.4)$$

$$P_{n+1}(t) = (2nt + 1)P_n(t) - t(t - 1)P'_n(t), \quad (18.5)$$

$$P_n(1) = (2n - 1)!! \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n - 1$ gaps of a Stirling permutation of order $n - 1$. A descent gap or the terminal gap preserves the number of descents, giving $k + 1$ choices from an object with k descents. Each of the remaining $2n - k - 1$ gaps creates one descent from an object with $k - 1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n - 1$ insertion gaps, so the total count is $(2n - 1)!!$. $\square$

The differential recurrence has two useful equivalent forms:

$$(t - 1)^{-2n-2} P_{n+1}(t) = (t(1 - t)^{-2n-1} P_n(t))', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1 - t} u_n(t) \right)', \quad u_n(t) = (t - 1)^{-2n} P_n(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.3 (Diagonal Stirling generating function). For $n \geq 1$,

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n + m \\ m \end{matrix} \right\} t^m = \frac{t P_n(t)}{(1 - t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1 - t)F_{n+1} = tF'_n$, with $F_0 = (1 - t)^{-1}$. Hence $F_1 = t/(1 - t)^3$. If the displayed formula holds for n , differentiating $tP_n/(1 - t)^{2n+1}$ and using (18.5) gives the formula for $n + 1$. $\square$

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.4 (Second-order Eulerian diagonal formula). For $p, m \geq 0$,

$$\left[\begin{matrix} m \\ m - p \end{matrix} \right] = \sum_{j=0}^p \langle\langle p \rangle\rangle_j \binom{m + j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\begin{bmatrix} x \\ x-p \end{bmatrix} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \begin{bmatrix} x \\ x-1 \end{bmatrix} &= \binom{x}{2}, \\ \begin{bmatrix} x \\ x-2 \end{bmatrix} &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \begin{bmatrix} x \\ x-3 \end{bmatrix} &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \begin{bmatrix} m \\ m-p \end{bmatrix} t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} P_p(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2 ((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} P_p(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $P_p(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Differentiation, Composition, and Reversion

Chapter 19

Faà di Bruno's formula

19.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 19.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (19.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

19.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 19.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (19.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (19.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (19.1) gives (19.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (19.3). $\square$

For $n = 4$ this reads

$$\begin{aligned} (f \circ g)^{(4)} &= f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 \\ &\quad + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \end{aligned} \quad (19.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

19.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (19.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (19.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

19.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial / \partial x_j$.

Theorem 19.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (19.6)$$

Proof. The proof of Theorem 19.1 applies verbatim, except that the new derivative is $\partial / \partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (19.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

Chapter 20

Composition of ordinary and exponential series

20.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 20.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \dots b_{i_k}, \quad (20.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \dots b_n^{\pi_n} \quad (20.2)$$

$$= \sum_{k=1}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad (20.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \widehat{\mathcal{B}}_{n,k}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (20.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

20.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 20.2 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}). \quad (20.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (20.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp \left(\sum_{j \geq 1} a_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (20.7)$$

If the series converge, (20.5) is exactly (19.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 21

Lagrange inversion and Lagrange–Bürmann

21.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (21.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 21.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n]H(g(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n, \quad (21.2)$$

$$= [w^n]H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (21.3)$$

In particular,

$$[z^n]g(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n. \quad (21.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.2 give

$$\begin{aligned} [z^n]H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1}f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (21.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (21.3). Take $H(w) = w$ for (21.4). □

21.2 Analytic inverse functions

Theorem 21.2 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (21.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (21.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta = z - b$, and

$$\phi(u) = \frac{u}{f(a + u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta \phi(u)$. Formula (21.4), multiplied by $n!$, is exactly (21.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + O((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

21.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$\hat{f}_j = \frac{f_{j+1}}{(j+1)f_1}. \quad (21.7)$$

Theorem 21.3 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k n^{\bar{k}} \mathcal{B}_{n-1,k}(\hat{f}_1, \dots, \hat{f}_{n-k}). \quad (21.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} \hat{f}_j \frac{w^j}{j!}.$$

By (21.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1 + U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

21.4 The associahedron interpretation, with normalized notation

The formula in the archive mixes exponential coefficients with an ordinary-series face formula. The exact geometric statement is the following.

Let

$$A(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad B(x) = A^{\langle -1 \rangle}(x) = x + \sum_{n \geq 1} b_n x^{n+1}. \quad (21.9)$$

Use the indexing in which the associahedron K_n has dimension $n - 1$ and its faces are plane rooted trees with $n + 1$ leaves and internal arities at least 2. If an internal vertex has arity $i + 1$, give it weight a_i ; let a_F be the product of all vertex weights. Equivalently, if $F \cong K_{i_1} \times \cdots \times K_{i_m}$, then $a_F = a_{i_1} \cdots a_{i_m}$.

Theorem 21.4 (Associahedral inversion).

$$b_n = \sum_{F \text{ a face of } K_n} (-1)^{n - \dim F} a_F. \quad (21.10)$$

Proof. The equation $A(B) = x$ is

$$B = x - \sum_{i \geq 1} a_i B^{i+1}.$$

Iterating it recursively expands b_n as a sum over plane rooted trees with $n + 1$ leaves: every internal vertex of arity $i + 1$ contributes $-a_i$. A face F of K_n is exactly such a tree after contracting a chosen set of internal edges. If it has m internal vertices, then $\dim F = n - m$, so its sign is $(-1)^m = (-1)^{n - \dim F}$ and its weight is a_F . Thus the tree expansion is the alternating face sum. $\square$

For example,

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_2a_1 - 5a_1^3, \\ b_4 &= -a_4 + 6a_3a_1 + 3a_2^2 - 21a_2a_1^2 + 14a_1^4. \end{aligned}$$

For K_4 , the coefficients 1, 6, 3, 21, 14 count respectively the whole 3-polytope, its six pentagonal and three square facets, its edges, and its vertices.

Chapter 22

Algebraic and transcendental examples

22.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (22.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 22.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (22.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (22.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (21.4),

$$[z^n]x(z) = \frac{1}{n}[w^{n-1}](1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

22.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 22.2. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + O(z^5), \quad (22.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (21.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

22.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (22.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + O(\xi^6). \quad (22.6)$$

Proof. The derivative at $u = 0$ of the left side of (22.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2$, $c_2 = 1/16$, $c_3 = -1/192$, $c_4 = -1/3072$, $c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143\dots$

22.4 Binary trees and Catalan numbers

Let C_n count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (22.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n = [z^n]C(z) = \frac{1}{n}[w^{n-1}](1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (22.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n is used here to avoid confusing these binary-tree numbers with Bell numbers.

Chapter 23

Lagrange's reversion theorem in shifted form

23.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (23.1)$$

Theorem 23.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (23.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (23.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (21.2) to $H(u) = g(x + u)$:

$$[y^k]g(v) = \frac{1}{k} [u^{k-1}]g'(x + u)f(x + u)^k = \frac{1}{k!} D_x^{k-1}(g'(x)f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

23.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x)g(z)(1 - yf'(z)) \, dz. \quad (23.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (23.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (23.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (23.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 24

Laplace-type asymptotic integrals

24.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (24.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (24.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (24.3)$$

Let $\widehat{\mathcal{B}}_{k,j}$ denote the ordinary Bell polynomial.

Theorem 24.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (24.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (24.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = tA(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s))t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} [t^k] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (21.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k]A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (24.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 24.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VII

Incidence, Umbral, and Matrix Methods

Chapter 25

Inversion on finite sets and partition lattices

The five drafts use several inversions that look different only because they are written in different coordinates. Binomial inversion is Möbius inversion on the Boolean lattice, Stirling inversion is change of basis between powers and factorial powers, and moment–cumulant inversion is Möbius inversion on the partition lattice. This chapter puts these statements in one framework and supplies the details that were abbreviated in the drafts.

25.1 Binomial inversion

Theorem 25.1 (Binomial inversion). *For sequences $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$ over a commutative ring,*

$$b_n = \sum_{k=0}^n \binom{n}{k} a_k \quad (25.1)$$

for every n if and only if

$$a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k. \quad (25.2)$$

Equivalently, if $A(z) = \sum_{n \geq 0} a_n z^n / n!$ and $B(z) = \sum_{n \geq 0} b_n z^n / n!$, then $B(z) = e^z A(z)$ and $A(z) = e^{-z} B(z)$.

Proof. Substitute (25.1) into the right side of (25.2) and interchange the two finite sums:

$$\begin{aligned} \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k &= \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} \sum_{j=0}^k \binom{k}{j} a_j \\ &= \sum_{j=0}^n a_j \binom{n}{j} \sum_{k=j}^n (-1)^{n-k} \binom{n-j}{k-j}. \end{aligned}$$

The inner sum is $(1-1)^{n-j}$, hence it is 1 when $j = n$ and 0 otherwise. The result is a_n . The reverse implication follows by the same calculation with the roles of the two formulas interchanged. For the generating functions, multiplication by e^z gives

$$n! [z^n] e^z A(z) = \sum_{k=0}^n \binom{n}{k} a_k = b_n,$$

and multiplication by e^{-z} gives the inverse relation. □

Remark 25.2. The sign $(-1)^{n-k}$ is the Möbius function of the Boolean interval from a k -element subset to an n -element set. Thus the theorem is the simplest instance of incidence-algebra inversion.

25.2 The Möbius function of the partition lattice

Let Π_n be the lattice of set partitions of $[n]$, ordered by refinement. For $\sigma \leq \pi$, and for each block B of π , let m_B be the number of blocks of σ contained in B .

Theorem 25.3 (Partition-lattice Möbius function). *The Möbius function of Π_n is*

$$\mu(\sigma, \pi) = (-1)^{|\sigma| - |\pi|} \prod_{B \in \pi} (m_B - 1)!. \quad (25.3)$$

In particular,

$$\mu(\pi, \hat{1}) = (-1)^{|\pi| - 1} (|\pi| - 1)!. \quad (25.4)$$

Proof. The interval $[\sigma, \pi]$ is the direct product of the lattices Π_{m_B} , one for each block B of π : passing from σ to a coarser partition inside B means partitioning the m_B blocks of σ that lie in B . The Möbius function is multiplicative on direct products, so it remains to compute

$$u_m := \mu_{\Pi_m}(\hat{0}, \hat{1}).$$

For a partition $\tau \in \Pi_m$, the interval $[\hat{0}, \tau]$ is the product of $\Pi_{|C|}$ over the blocks C of τ . Hence

$$\mu(\hat{0}, \tau) = \prod_{C \in \tau} u_{|C|}.$$

The defining Möbius relation says

$$\sum_{\tau \in \Pi_m} \prod_{C \in \tau} u_{|C|} = \begin{cases} 1, & m = 0, 1, \\ 0, & m \geq 2, \end{cases}$$

where the empty partition contributes 1 for $m = 0$. By the labelled exponential formula, the exponential generating function of the left side is

$$\exp \left(\sum_{m \geq 1} u_m \frac{z^m}{m!} \right).$$

The displayed values therefore force this exponential to equal $1 + z$. Taking the formal logarithm gives

$$\sum_{m \geq 1} u_m \frac{z^m}{m!} = \log(1 + z) = \sum_{m \geq 1} (-1)^{m-1} \frac{z^m}{m},$$

so $u_m = (-1)^{m-1} (m-1)!$. Multiplication over the blocks of π yields (25.3); taking $\pi = \hat{1}$ yields (25.4). $\square$

25.3 Moments and cumulants as Möbius transforms

Let $(m_n)_{n \geq 0}$ and $(\kappa_n)_{n \geq 1}$ be related by

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad m_0 = 1. \quad (25.5)$$

This is the set-partition form of the moment–cumulant relation.

Corollary 25.4 (Cumulant inversion). *The inverse relation is*

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|}. \quad (25.6)$$

Equivalently, if

$$M(z) = 1 + \sum_{n \geq 1} m_n \frac{z^n}{n!}, \quad K(z) = \sum_{n \geq 1} \kappa_n \frac{z^n}{n!},$$

then $M(z) = \exp K(z)$ and $K(z) = \log M(z)$.

Proof. Equation (25.5) says that the multiplicative function built from the m_n is the zeta transform of the multiplicative function built from the κ_n on the partition lattice. Möbius inversion and (25.4) give (25.6). Grouping partitions by their block-size multiplicities gives the complete Bell-polynomial form, and the exponential formula gives $M = \exp K$. Taking the formal logarithm gives the inverse. $\square$

Chapter 26

Binomial type, Sheffer sequences, and exponential Riordan arrays

26.1 Polynomial sequences of binomial type

Definition 26.1. A sequence $(p_n(x))_{n \geq 0}$ with $p_0(x) = 1$ and $\deg p_n = n$ is of *binomial type* if

$$p_n(x + y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (26.1)$$

A formal series $\beta(z) = \beta_1 z + \beta_2 z^2 + \cdots$ with $\beta_1 \neq 0$ is called a delta series.

Theorem 26.2 (Delta-series characterization). *Over a characteristic-zero field, (p_n) is of binomial type if and only if there is a unique delta series β such that*

$$\sum_{n \geq 0} p_n(x) \frac{z^n}{n!} = \exp(x\beta(z)). \quad (26.2)$$

For this series,

$$p_n(x) = \sum_{k=0}^n x^k \mathcal{B}_{n,k}(\beta_1, \beta_2, \dots, \beta_{n-k+1}). \quad (26.3)$$

Proof. If (26.2) holds, then the product of the generating functions at x and y is the generating function at $x + y$; coefficient comparison gives (26.1).

Conversely, put $P(x, z) = \sum p_n(x) z^n / n!$. The binomial identity is exactly $P(x + y, z) = P(x, z)P(y, z)$, while $P(0, z) = 1$. Hence the formal logarithm $L(x, z) = \log P(x, z)$ satisfies $L(x + y, z) = L(x, z) + L(y, z)$. For each fixed power of z , its coefficient is a polynomial in x that is additive. In characteristic zero every additive polynomial is linear, so $L(x, z) = x\beta(z)$. The degree condition forces $\beta(0) = 0$ and $\beta_1 \neq 0$, and uniqueness is clear from $\beta(z) = L(1, z)$. Finally, expand

$$e^{x\beta(z)} = \sum_{k \geq 0} \frac{x^k}{k!} \beta(z)^k$$

and use the defining exponential generating function of the partial Bell polynomials. This gives (26.3). $\square$

Example 26.3. The choices $\beta(z) = z, \log(1 + z), -\log(1 - z)$, and $e^z - 1$ give, respectively,

$$x^n, \quad x^{\overline{n}}, \quad x^{\overline{n}}, \quad T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

Thus monomials, factorial powers, and Touchard polynomials are coordinates in one delta-series calculus.

26.2 Sheffer sequences and lowering operators

Definition 26.4. A polynomial sequence $(s_n(x))_{n \geq 0}$ is a *Sheffer sequence* for (A, β) if

$$\sum_{n \geq 0} s_n(x) \frac{z^n}{n!} = A(z) e^{x\beta(z)}, \quad A(0) \neq 0, \quad (26.4)$$

where β is a delta series.

Let (p_n) be the binomial-type sequence associated with β and write $A(z) = \sum a_j z^j / j!$.

Proposition 26.5 (Sheffer addition and convolution). *One has*

$$s_n(x+y) = \sum_{k=0}^n \binom{n}{k} s_k(x) p_{n-k}(y), \quad (26.5)$$

$$s_n(x) = \sum_{j=0}^n \binom{n}{j} a_j p_{n-j}(x). \quad (26.6)$$

If $\bar{\beta}$ is the compositional inverse of β and $Q = \bar{\beta}(D_x)$, then

$$Q p_n(x) = n p_{n-1}(x). \quad (26.7)$$

Proof. Multiplying the EGF at x by $e^{y\beta(z)}$ gives the EGF at $x+y$ and proves (26.5). Multiplication of the EGFs of A and p_n gives (26.6). Finally, $D_x e^{x\beta(z)} = \beta(z) e^{x\beta(z)}$. Formal functional calculus therefore gives

$$\bar{\beta}(D_x) e^{x\beta(z)} = \bar{\beta}(\beta(z)) e^{x\beta(z)} = z e^{x\beta(z)}.$$

Comparison of coefficients proves (26.7). □

Remark 26.6. Appell sequences are the special case $\beta(z) = z$. The Bernoulli and Euler polynomials developed later are Appell sequences with prefactors $t/(e^t - 1)$ and $2/(e^t + 1)$, respectively. Lagrange inversion enters the operational theory through the coefficients of $\bar{\beta}$.

26.3 Exponential Riordan arrays

For $g(0) \neq 0$ and $f(z) = f_1 z + O(z^2)$ with $f_1 \neq 0$, define

$$R_{n,k} = \frac{n!}{k!} [z^n] g(z) f(z)^k. \quad (26.8)$$

Theorem 26.7 (Riordan action, product, and inverse). *If (a_k) has EGF $A(z)$, the transformed sequence $b_n = \sum_{k=0}^n R_{n,k} a_k$ has EGF*

$$B(z) = g(z) A(f(z)). \quad (26.9)$$

Consequently,

$$(g, f)(h, \ell) = (g h \circ f, \ell \circ f), \quad (26.10)$$

$$(g, f)^{-1} = \left(\frac{1}{g \circ \bar{f}}, \bar{f} \right), \quad (26.11)$$

where $\bar{f} = f^{\langle -1 \rangle}$.

Proof. Using (26.8),

$$\begin{aligned} \sum_{n \geq 0} b_n \frac{z^n}{n!} &= \sum_{k \geq 0} a_k \sum_{n \geq k} \frac{R_{n,k}}{n!} z^n \\ &= g(z) \sum_{k \geq 0} a_k \frac{f(z)^k}{k!} = g(z) A(f(z)). \end{aligned}$$

Applying first (h, ℓ) and then (g, f) sends A to $g(z)h(f(z))A(\ell(f(z)))$, proving the product law. Substituting $\bar{f}$ into the product and choosing the prefactor $1/(g \circ \bar{f})$ gives the identity array $(1, z)$, proving the inverse formula. $\square$

The principal coefficient arrays of this monograph are Riordan arrays:

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{k!} [z^n] (e^z - 1)^k, \quad (26.12)$$

$$s(n, k) = \frac{n!}{k!} [z^n] \log(1 + z)^k, \quad (26.13)$$

$$\mathcal{B}_{n,k}(x_1, x_2, \dots) = \frac{n!}{k!} [z^n] \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right)^k. \quad (26.14)$$

Since $e^z - 1$ and $\log(1 + z)$ are compositional inverses, Stirling matrix inversion is the special case of (26.11) with $g = 1$. Faà di Bruno composition is matrix multiplication in the same group.

26.4 Abel polynomials from Lagrange inversion

Theorem 26.8 (Abel binomial sequence). *For a fixed scalar a , the polynomials*

$$A_n(x) = x(x - an)^{n-1} \quad (n \geq 1), \quad A_0(x) = 1, \quad (26.15)$$

form a sequence of binomial type. Hence

$$(x + y)(x + y - an)^{n-1} = \sum_{k=0}^n \binom{n}{k} x(x - ak)^{k-1} y(y - a(n - k))^{n-k-1}, \quad (26.16)$$

with the endpoint factors interpreted by continuity as $A_0 = 1$.

Proof. Let $T(z)$ be the unique formal solution of $T = ze^{-aT}$. Lagrange–Bürmann inversion gives, for $n \geq 1$,

$$\begin{aligned} [z^n] e^{xT(z)} &= \frac{x}{n} [u^{n-1}] e^{xu} e^{-anu} \\ &= \frac{x(x - an)^{n-1}}{n!}. \end{aligned}$$

Therefore

$$e^{xT(z)} = \sum_{n \geq 0} A_n(x) \frac{z^n}{n!}.$$

This is of the form (26.2); the delta-series characterization proves binomial type, and coefficient comparison in $e^{(x+y)T} = e^{xT} e^{yT}$ gives (26.16). $\square$

Chapter 27

Triangular algorithms and independent consistency checks

27.1 Elementary operations on ordinary series

Let $A(z) = \sum_{n \geq 0} a_n z^n$, $B(z) = \sum_{n \geq 0} b_n z^n$, and $C(z) = \sum_{n \geq 0} c_n z^n$. The following formulas are triangular: once coefficients of order below n are known, they determine the coefficient of order n .

Proposition 27.1 (Product, reciprocal, and quotient). *For $C = AB$,*

$$c_n = \sum_{j=0}^n a_j b_{n-j}. \quad (27.1)$$

If $a_0 \neq 0$ and $C = 1/A$, then

$$c_0 = a_0^{-1}, \quad c_n = -a_0^{-1} \sum_{j=1}^n a_j c_{n-j}. \quad (27.2)$$

If $C = B/A$, then

$$c_n = a_0^{-1} \left(b_n - \sum_{j=1}^n a_j c_{n-j} \right). \quad (27.3)$$

Proof. All three statements are obtained by equating the coefficient of z^n in $C = AB$, $AC = 1$, and $AC = B$, respectively, and isolating the term containing c_n . $\square$

Proposition 27.2 (Logarithm, exponential, and arbitrary powers). *Assume $a_0 = 1$. If $L = \log A = \sum_{n \geq 1} \ell_n z^n$, then*

$$\ell_n = a_n - \frac{1}{n} \sum_{j=1}^{n-1} j \ell_j a_{n-j}. \quad (27.4)$$

Conversely, if $A = e^L$, then

$$a_0 = 1, \quad a_n = \frac{1}{n} \sum_{j=1}^n j \ell_j a_{n-j}. \quad (27.5)$$

For $C = A^\alpha$ with $c_0 = 1$,

$$c_n = \frac{1}{n} \sum_{j=1}^n ((\alpha + 1)j - n) a_j c_{n-j}. \quad (27.6)$$

Proof. The identity $A' = AL'$ gives $na_n = \sum_{j=1}^n j\ell_j a_{n-j}$; solving either for ℓ_n or for a_n proves the first two recurrences. For the power, logarithmic differentiation gives $AC' = \alpha A'C$. The coefficient of z^{n-1} is

$$\sum_{j=0}^{n-1} (n-j)a_j c_{n-j} = \alpha \sum_{j=1}^n j a_j c_{n-j}.$$

Isolating nc_n gives (27.6); its $j = n$ term is αa_n . □

27.2 Composition and reversion

If $B(0) = 0$, the ordinary Bell polynomials give

$$[z^n]A(B(z)) = \sum_{k=0}^n a_k [z^n]B(z)^k. \quad (27.7)$$

The coefficients of each power B^k follow from repeated Cauchy products, or from the ordinary partial Bell-polynomial recurrence. If $F(z) = a_1 z + a_2 z^2 + \cdots$ with $a_1 \neq 0$, its inverse G is determined by

$$[z^n]G(z)^m = \frac{m}{n}[u^{n-m}] \left(\frac{u}{F(u)} \right)^n, \quad 1 \leq m \leq n, \quad (27.8)$$

which is the power form of Lagrange inversion. Thus product, quotient, powers, logarithm, exponential, composition, and reversion all reduce to finite convolution and coefficient extraction.

27.3 Low-order inverse certificates

Proposition 27.3 (Direct reversion through degree five). *For*

$$F(z) = z + a_2 z^2 + a_3 z^3 + a_4 z^4 + a_5 z^5 + O(z^6),$$

the compositional inverse is

$$\begin{aligned} F^{(-1)}(z) = & z - a_2 z^2 + (2a_2^2 - a_3)z^3 \\ & + (-5a_2^3 + 5a_2 a_3 - a_4)z^4 \\ & + (14a_2^4 - 21a_2^2 a_3 + 3a_3^2 + 6a_2 a_4 - a_5)z^5 + O(z^6). \end{aligned}$$

Proof. Write $G(z) = z + b_2 z^2 + b_3 z^3 + b_4 z^4 + b_5 z^5 + O(z^6)$ and expand $F(G(z))$. The coefficients of z^2, z^3, z^4, z^5 are respectively

$$\begin{aligned} & b_2 + a_2, \\ & b_3 + 2a_2 b_2 + a_3, \\ & b_4 + a_2(2b_3 + b_2^2) + 3a_3 b_2 + a_4, \\ & b_5 + a_2(2b_4 + 2b_2 b_3) + a_3(3b_3 + 3b_2^2) + 4a_4 b_2 + a_5. \end{aligned}$$

Setting them equal to zero and solving successively gives the displayed coefficients. This finite calculation independently certifies the Bell- and Lagrange-based inverse formulas through fourth nontrivial order. □

Part VIII

Bernoulli, Euler, Genocchi, and Summation Calculus

Chapter 28

Bernoulli polynomials and their two kinds

28.1 Definition and Appell structure

The Bernoulli polynomials $\beta_n(x)$ are defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} \beta_n(x) \frac{t^n}{n!}. \quad (28.1)$$

Thus $\beta_n = \beta_n(0)$, with $\beta_1 = -1/2$. The identity is valid formally in t and analytically for $|t| < 2\pi$. These normalizations and the parallel Euler-polynomial conventions agree with the authoritative tables in [13]; the finite-difference viewpoint is classical in [18].

Theorem 28.1 (Basic Bernoulli-polynomial calculus). *For $n \geq 0$,*

$$\beta'_n(x) = n\beta_{n-1}(x), \quad n \geq 1, \quad (28.2)$$

$$\beta_n(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k(x) y^{n-k}, \quad (28.3)$$

$$\beta_n(x+1) - \beta_n(x) = nx^{n-1}, \quad n \geq 1, \quad (28.4)$$

$$\beta_n(1-x) = (-1)^n \beta_n(x), \quad (28.5)$$

$$\int_0^1 \beta_n(x) dx = 0, \quad n \geq 1. \quad (28.6)$$

For every positive integer q ,

$$\beta_n(qx) = q^{n-1} \sum_{r=0}^{q-1} \beta_n\left(x + \frac{r}{q}\right). \quad (28.7)$$

Proof. Differentiate (28.1) with respect to x and compare coefficients to obtain (28.2). Multiplication of the EGF by e^{yt} gives (28.3). For the finite difference,

$$\frac{te^{(x+1)t}}{e^t - 1} - \frac{te^{xt}}{e^t - 1} = te^{xt},$$

and coefficient comparison gives (28.4). For reflection,

$$\frac{te^{(1-x)t}}{e^t - 1} = \frac{(-t)e^{x(-t)}}{e^{-t} - 1},$$

which is the EGF at x with t replaced by $-t$.

Integrating (28.2) from 0 to 1 gives

$$n \int_0^1 \beta_{n-1}(x) dx = \beta_n(1) - \beta_n(0).$$

The right side is zero for $n \neq 1$ by (28.4); the remaining case is $\int_0^1 \beta_0(x) dx = 1$, so after reindexing (28.6) follows for positive degree.

For the multiplication theorem, sum the EGFs:

$$\begin{aligned} \sum_{r=0}^{q-1} \frac{te^{(x+r/q)t}}{e^t - 1} &= \frac{te^{xt}}{e^t - 1} \frac{e^t - 1}{e^{t/q} - 1} \\ &= \frac{te^{xt}}{e^{t/q} - 1}. \end{aligned}$$

Replace t by qt and compare coefficients. This gives $\sum_r \beta_n(x + r/q) = q^{1-n} \beta_n(qx)$, equivalent to (28.7). $\square$

28.2 Faulhaber's formula and a Stirling expansion

Theorem 28.2 (Faulhaber's formula). *For integers $N \geq 1$ and $p \geq 0$,*

$$\sum_{k=0}^{N-1} k^p = \frac{\beta_{p+1}(N) - \beta_{p+1}}{p+1}. \quad (28.8)$$

Equivalently,

$$\sum_{k=0}^{N-1} k^p = \frac{1}{p+1} \sum_{j=0}^p \binom{p+1}{j} \beta_j N^{p+1-j}. \quad (28.9)$$

Proof. Apply (28.4) with $n = p+1$ and $x = k$:

$$\beta_{p+1}(k+1) - \beta_{p+1}(k) = (p+1)k^p.$$

Summation over $k = 0, \dots, N-1$ telescopes and gives (28.8). The addition formula at $x = 0, y = N$ expands $\beta_{p+1}(N)$ and gives (28.9); the term $j = p+1$ cancels β_{p+1} . $\square$

Theorem 28.3 (Bernoulli numbers from Stirling numbers). *For every $n \geq 0$,*

$$\beta_n = \sum_{k=0}^n (-1)^k \frac{k!}{k+1} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (28.10)$$

Proof. Put $y = e^t - 1$. Since $t = \log(1+y)$,

$$\frac{t}{e^t - 1} = \frac{\log(1+y)}{y} = \sum_{k \geq 0} \frac{(-1)^k}{k+1} y^k.$$

The Stirling EGF is

$$(e^t - 1)^k = k! \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{t^n}{n!}.$$

Substitute this into the preceding series and compare the coefficient of $t^n/n!$. $\square$

28.3 Two modified conventions

Some references replace only the exceptional value $\beta_1 = -1/2$ by $+1/2$. Define β_n^+ by

$$\frac{te^t}{e^t - 1} = \sum_{n \geq 0} \beta_n^+ \frac{t^n}{n!}. \quad (28.11)$$

Then $\beta_1^+ = 1/2$ and $\beta_n^+ = \beta_n$ for $n \neq 1$. This is a notational convention, not a new sequence in positive even degree.

A different object, also called a modified Bernoulli sequence in the supplied MathWorld material, is encoded by the logarithm of a hyperbolic sine.

Proposition 28.4 (Modified Bernoulli logarithm). *For $|z| < 2\pi$,*

$$\frac{1}{2} \log \left(\frac{e^{z/2} - e^{-z/2}}{z/2} \right) = \frac{1}{2} \log 2 + \sum_{m \geq 1} \frac{\beta_{2m}}{4m(2m)!} z^{2m}. \quad (28.12)$$

Proof. Differentiate the left side:

$$\frac{1}{4} \coth \frac{z}{2} - \frac{1}{2z}.$$

From the Bernoulli generating function,

$$\coth \frac{z}{2} = \frac{2}{z} + 2 \sum_{m \geq 1} \beta_{2m} \frac{z^{2m-1}}{(2m)!}.$$

The singular terms cancel, leaving $\sum_{m \geq 1} \beta_{2m} z^{2m-1} / (2(2m)!)$. Integration gives the series, and the value at $z = 0$ is $\frac{1}{2} \log 2$. $\square$

28.4 Bernoulli polynomials of the second kind

Define the Bernoulli polynomials of the second kind, also called Cauchy polynomials in this normalization, by

$$\frac{z(1+z)^x}{\log(1+z)} = \sum_{n \geq 0} \mathfrak{b}_n(x) \frac{z^n}{n!}. \quad (28.13)$$

Their values $\mathfrak{b}_n = \mathfrak{b}_n(0)$ begin

$$1, \quad \frac{1}{2}, \quad -\frac{1}{6}, \quad \frac{1}{4}, \quad -\frac{19}{30}, \dots$$

Theorem 28.5 (Integral and difference representations). *For every $n \geq 0$,*

$$\mathfrak{b}_n(x) = \int_x^{x+1} u^n du, \quad (28.14)$$

$$\mathfrak{b}'_n(x) = nx^{n-1}, \quad n \geq 1, \quad (28.15)$$

$$\mathfrak{b}_n(x+1) - \mathfrak{b}_n(x) = n\mathfrak{b}_{n-1}(x), \quad n \geq 1. \quad (28.16)$$

At $x = 0$,

$$\boxed{\mathfrak{b}_n = \sum_{k=0}^n \frac{s(n, k)}{k+1}}. \quad (28.17)$$

Proof. Since

$$\int_x^{x+1} (1+z)^u \, du = \frac{z(1+z)^x}{\log(1+z)},$$

and $(1+z)^u = \sum_{n \geq 0} u^n z^n / n!$, coefficient comparison gives (28.14). Differentiating the integral with moving endpoints gives

$$x + 1^n - x^n = nx^{n-1},$$

which proves (28.15). Alternatively, differentiating the EGF with respect to x cancels the logarithm and gives the same result. Replacing x by $x+1$ in the EGF multiplies it by $1+z$; subtracting the original EGF therefore multiplies it by z , proving (28.16). Finally, $u^n = \sum_{k=0}^n s(n, k) u^k$; integrate from 0 to 1 to obtain (28.17). $\square$

Chapter 29

Euler polynomials, Genocchi numbers, and alternating sums

29.1 Euler polynomials

Define the Euler polynomials $\mathcal{E}_n(x)$ by

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} \mathcal{E}_n(x) \frac{t^n}{n!}. \quad (29.1)$$

This series is formal and is analytic for $|t| < \pi$.

Theorem 29.1 (Euler-polynomial calculus). *For $n \geq 0$,*

$$\mathcal{E}'_n(x) = n\mathcal{E}_{n-1}(x), \quad n \geq 1, \quad (29.2)$$

$$\mathcal{E}_n(x+y) = \sum_{k=0}^n \binom{n}{k} \mathcal{E}_k(x) y^{n-k}, \quad (29.3)$$

$$\mathcal{E}_n(x+1) + \mathcal{E}_n(x) = 2x^n, \quad (29.4)$$

$$\mathcal{E}_n(1-x) = (-1)^n \mathcal{E}_n(x). \quad (29.5)$$

Moreover,

$$\boxed{\mathcal{E}_n(x) = \frac{2^{n+1}}{n+1} \left[\beta_{n+1} \left(\frac{x+1}{2} \right) - \beta_{n+1} \left(\frac{x}{2} \right) \right]}. \quad (29.6)$$

Proof. Differentiation in x and multiplication by e^{yt} prove the first two identities. The sum of the EGFs at $x+1$ and x is $2e^{xt}$, proving (29.4). Also,

$$\frac{2e^{(1-x)t}}{e^t + 1} = \frac{2e^{x(-t)}}{e^{-t} + 1},$$

which proves reflection.

For the Bernoulli relation, form the EGF of its right side. With $m = n+1$, it is

$$\begin{aligned} \frac{1}{t} \left(\sum_{m \geq 0} \beta_m \left(\frac{x+1}{2} \right) \frac{(2t)^m}{m!} - \sum_{m \geq 0} \beta_m \left(\frac{x}{2} \right) \frac{(2t)^m}{m!} \right) &= \frac{2(e^{(x+1)t} - e^{xt})}{e^{2t} - 1} \\ &= \frac{2e^{xt}}{e^t + 1}. \end{aligned}$$

The constant terms cancel before division by t , so coefficient comparison is valid and proves (29.6). $\square$

Corollary 29.2 (Finite alternating power sums). *For $N \geq 1$,*

$$\sum_{k=0}^{N-1} (-1)^k (x+k)^n = \frac{\mathcal{E}_n(x) + (-1)^{N-1} \mathcal{E}_n(x+N)}{2}. \quad (29.7)$$

Proof. Replace x in (29.4) by $x+k$, multiply by $(-1)^k$, and sum. All interior Euler-polynomial terms cancel pairwise, leaving the two endpoints. $\square$

29.2 Genocchi numbers and polynomials

Define the Genocchi polynomials $G_n(x)$ by

$$\frac{2te^{xt}}{e^t + 1} = \sum_{n \geq 0} G_n(x) \frac{t^n}{n!}, \quad G_n = G_n(0). \quad (29.8)$$

Thus $G_0 = 0$, $G_1 = 1$, $G_2 = -1$, $G_4 = 1$, and $G_6 = -3$.

Theorem 29.3 (Genocchi relations). *For $n \geq 1$,*

$$G_n(x) = n\mathcal{E}_{n-1}(x), \quad (29.9)$$

$$G_n = 2(1 - 2^n)\beta_n, \quad (29.10)$$

$$G_n(x+1) + G_n(x) = 2nx^{n-1}. \quad (29.11)$$

In particular, $G_{2m+1} = 0$ for $m \geq 1$.

Proof. Multiplying the Euler-polynomial EGF by t shifts its coefficients and gives (29.9). At $x = 0$,

$$\begin{aligned} \frac{2t}{e^t + 1} &= \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1} \\ &= 2 \sum_{n \geq 0} \beta_n \frac{t^n}{n!} - 2 \sum_{n \geq 0} \beta_n \frac{(2t)^n}{n!}, \end{aligned}$$

which proves (29.10). Multiplying (29.4) by n and replacing n there by $n-1$ gives (29.11). The vanishing of positive odd indices above 1 follows from the corresponding Bernoulli-number vanishing. $\square$

If $\tan z = \sum_{m \geq 1} T_{2m-1} z^{2m-1} / (2m-1)!$, then

$$T_{2m-1} = (-1)^m \frac{2^{2m-2}}{m} G_{2m}. \quad (29.12)$$

Indeed, the standard identity $T_{2m-1} = (-1)^{m-1} 2^{2m} (2^{2m} - 1) \beta_{2m} / (2m)$ and (29.10) are algebraically equivalent; alternatively it follows by comparing the generating functions of $\tan z$ and the even part of $2t/(e^t + 1)$.

Chapter 30

Complementary Bell numbers and reciprocal partition transforms

Let $T_n(x) = \sum_{k=0}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^k$ be the Touchard polynomial. The complementary Bell number is

$$C_n := T_n(-1) = \sum_{k=0}^n (-1)^k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}. \quad (30.1)$$

It is the signed enumeration of set partitions, where a partition with k blocks has weight $(-1)^k$.

Theorem 30.1 (Generating function, Dobinski formula, and recurrence). *The complementary Bell numbers satisfy*

$$\sum_{n \geq 0} C_n \frac{z^n}{n!} = \exp(1 - e^z), \quad (30.2)$$

$$C_n = e \sum_{m \geq 0} \frac{(-1)^m m^n}{m!}, \quad (30.3)$$

$$C_{n+1} = - \sum_{k=0}^n \binom{n}{k} C_k. \quad (30.4)$$

The first values are

$$1, -1, 0, 1, 1, -2, -9, -9, 50, 267, 413, \dots$$

Proof. The Touchard EGF is $\exp(x(e^z - 1))$; setting $x = -1$ gives (30.2). Expanding $\exp(-e^z)$ and extracting coefficients gives

$$e \exp(-e^z) = e \sum_{m \geq 0} \frac{(-1)^m e^{mz}}{m!},$$

which proves (30.3). If $C(z)$ denotes the EGF in (30.2), then $C'(z) = -e^z C(z)$. Exponential coefficient comparison gives (30.4). $\square$

Corollary 30.2 (Bell and complementary Bell inversion). *For every $n \geq 0$,*

$$\sum_{k=0}^n \binom{n}{k} B_k C_{n-k} = \delta_{n0}. \quad (30.5)$$

Proof. The Bell EGF is $\exp(e^z - 1)$, the reciprocal of (30.2). The coefficient of $z^n/n!$ in their product is the left side. $\square$

More generally, $T_n(-x)$ is the signed block enumerator with a weight $-x$ per block and EGF $\exp(-x(e^z - 1))$. This one-parameter form clarifies that the complementary sequence is not an unrelated table but the inverse, in the exponential-convolution group, of the ordinary Bell sequence.

Chapter 31

Darboux formulas, Euler–Maclaurin summation, and coefficient transfer

31.1 Darboux’s generalized Taylor formula

The name “Darboux’s formula” is used for more than one result. The supplied MathWorld notebook records the following weighted Taylor identity. It should not be confused with the coefficient-asymptotic method proved later in this chapter.

Theorem 31.1 (Darboux’s weighted Taylor identity). *Let ϕ be a polynomial of exact degree n and let f have $n + 1$ continuous derivatives on the segment joining z_0 to z . Put $h = z - z_0$ and let $\phi^{(n)}$ denote the nonzero constant n th derivative. Then*

$$f(z) = f(z_0) + \frac{1}{\phi^{(n)}} \left\{ (-1)^n h^{n+1} \int_0^1 \phi(t) f^{(n+1)}(z_0 + th) dt + \sum_{k=1}^n (-1)^{k-1} h^k \left[\phi^{(n-k)}(1) f^{(k)}(z) - \phi^{(n-k)}(0) f^{(k)}(z_0) \right] \right\}. \quad (31.1)$$

Proof. Set $F(t) = f(z_0 + th)$, so $F^{(k)}(t) = h^k f^{(k)}(z_0 + th)$. Integrate $\int_0^1 \phi(t) F^{(n+1)}(t) dt$ by parts $n + 1$ times. Because $\phi^{(n+1)} = 0$,

$$\int_0^1 \phi F^{(n+1)} dt = \sum_{j=0}^n (-1)^j \left[\phi^{(j)} F^{(n-j)} \right]_0^1.$$

The term $j = n$ is $(-1)^n \phi^{(n)} (F(1) - F(0))$. Solve for $F(1) - F(0)$, replace j by $n - k$ in the remaining sum, and substitute the derivatives of F . This gives exactly (31.1). $\square$

Corollary 31.2 (Taylor’s integral remainder). *Taking $\phi(t) = (1 - t)^n$ gives*

$$f(z) = \sum_{k=0}^n \frac{f^{(k)}(z_0)}{k!} h^k + \frac{h^{n+1}}{n!} \int_0^1 (1 - t)^n f^{(n+1)}(z_0 + th) dt. \quad (31.2)$$

Proof. For $k \leq n$, the derivatives $\phi^{(n-k)}(1)$ vanish, while $\phi^{(n-k)}(0) = (-1)^{n-k} n! / k!$ and $\phi^{(n)} = (-1)^n n!$. Substitute these values into (31.1) and simplify signs. $\square$

31.2 Periodic Bernoulli functions

For $m \neq 1$, let $\tilde{\beta}_m(x) = \beta_m(\{x\})$, where $\{x\}$ is the fractional part. For $m = 1$, take the periodic sawtooth equal to $\{x\} - 1/2$ away from the integers; its values at the jump points do not affect

integrals. Between consecutive integers,

$$\frac{d}{dx} \tilde{\beta}_m(x) = m \tilde{\beta}_{m-1}(x). \quad (31.3)$$

Proposition 31.3 (Fourier series and uniform bound). *For $p \geq 1$,*

$$\tilde{\beta}_{2p}(x) = (-1)^{p+1} \frac{2(2p)!}{(2\pi)^{2p}} \sum_{m=1}^{\infty} \frac{\cos(2\pi m x)}{m^{2p}}. \quad (31.4)$$

Consequently,

$$|\tilde{\beta}_{2p}(x)| \leq \frac{2(2p)! \zeta(2p)}{(2\pi)^{2p}}. \quad (31.5)$$

Proof. For $p = 1$, compute the Fourier coefficients of $x^2 - x + 1/6$ on $[0, 1]$ by two integrations by parts. For general p , repeated integration by parts and (31.3) show that the m th nonzero Fourier coefficient is multiplied by $-(2p)(2p-1)/(2\pi m)^2$ when the index increases from $2p-2$ to $2p$. This induction gives (31.4). Absolute convergence for $p \geq 1$ and the triangle inequality give (31.5). $\square$

31.3 Euler–Maclaurin summation with remainder

Theorem 31.4 (Euler–Maclaurin). *Let $a < b$ be integers, let $p \geq 1$, and let $f \in C^{2p}([a, b])$. Then*

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} \\ &\quad + \sum_{r=1}^p \frac{\beta_{2r}}{(2r)!} \left(f^{(2r-1)}(b) - f^{(2r-1)}(a) \right) + R_p, \end{aligned} \quad (31.6)$$

where

$$R_p = -\frac{1}{(2p)!} \int_a^b \tilde{\beta}_{2p}(x) f^{(2p)}(x) dx. \quad (31.7)$$

Therefore

$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_a^b |f^{(2p)}(x)| dx. \quad (31.8)$$

Proof. On one unit interval $[k, k+1]$, integration by parts gives

$$\int_k^{k+1} \left(x - k - \frac{1}{2}\right) f'(x) dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x) dx.$$

Summing from $k = a$ to $b-1$ yields

$$\sum_{k=a}^b f(k) = \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \int_a^b \tilde{\beta}_1(x) f'(x) dx. \quad (31.9)$$

Integrate the last integral by parts piecewise on every unit interval, using (31.3). The interior endpoint terms cancel because $\beta_m(0) = \beta_m(1)$ for $m \neq 1$. Two integrations replace $\tilde{\beta}_{2r-2} f^{(2r-2)}$ by the boundary term involving β_{2r} and a new integral involving $-\tilde{\beta}_{2r} f^{(2r)}$. Iterating through $r = p$ proves (31.6) and (31.7). The bound follows immediately from (31.5). $\square$

Remark 31.5. When f is a polynomial of degree at most $2p-1$, the remainder vanishes and Euler–Maclaurin is exact. Applied to $f(x) = x^m$, it reproduces Faulhaber’s formula. Applied to $f(x) = \log x$, it produces Stirling’s asymptotic series; applied to $f(x) = x \log x$, it produces the Barnes G expansion developed later.

31.4 Darboux coefficient transfer

The second common use of Darboux’s name concerns coefficients controlled by a boundary singularity.

Theorem 31.6 (Darboux transfer with analytic multiplier). *Let $\rho > 0$, let H be analytic in $|z| < R$ for some $R > \rho$, and let $\alpha \in \mathbb{C}$. Suppose*

$$F(z) = (1 - z/\rho)^{-\alpha} H(z).$$

For a fixed positive integer M , put

$$h_j = \frac{(-\rho)^j}{j!} H^{(j)}(\rho). \quad (31.10)$$

Then

$$\begin{aligned} [z^n]F(z) &= \rho^{-n} \sum_{j=0}^{M-1} h_j \binom{n + \alpha - j - 1}{n} \\ &\quad + O\left(\rho^{-n} n^{\Re \alpha - M - 1}\right), \quad n \rightarrow \infty. \end{aligned} \quad (31.11)$$

The generalized binomial coefficient is interpreted by analytic continuation; when the upper parameter makes the term a polynomial, the same formula is understood in that limiting sense. In particular, if $H(\rho) \neq 0$ and $\alpha \notin \{0, -1, -2, \dots\}$,

$$[z^n]F(z) \sim \frac{H(\rho)}{\Gamma(\alpha)} \rho^{-n} n^{\alpha-1}. \quad (31.12)$$

Proof. Taylor-expand H at ρ in the coordinate $1 - z/\rho$:

$$H(z) = \sum_{j=0}^{M-1} h_j (1 - z/\rho)^j + (1 - z/\rho)^M H_M(z),$$

where H_M is analytic in a disk of radius greater than ρ . Multiplying by $(1 - z/\rho)^{-\alpha}$ gives the finite main sum in (31.11). The exact coefficient identity

$$[z^n](1 - z/\rho)^{-\gamma} = \rho^{-n} \binom{n + \gamma - 1}{n} = \rho^{-n} \frac{\Gamma(n + \gamma)}{\Gamma(\gamma)\Gamma(n + 1)}$$

provides each main term.

Write $H_M(z) = \sum_{k \geq 0} q_k z^k$. Cauchy’s estimate gives $|q_k| \leq C R_0^{-k}$ for some $R_0 > \rho$. Convolution with the coefficients of $(1 - z/\rho)^{M-\alpha}$ then gives

$$|[z^n](1 - z/\rho)^{M-\alpha} H_M(z)| \leq C' \rho^{-n} n^{\Re \alpha - M - 1} \sum_{k \geq 0} \left(\frac{\rho}{R_0}\right)^k (1 + k)^{C''},$$

where the finitely many terms near $k = n$ are even smaller because of the exponential factor. The convergent sum proves the error estimate. Finally, Stirling’s formula for the gamma ratio gives $\Gamma(n + \alpha)/\Gamma(n + 1) \sim n^{\alpha-1}$, proving (31.12). $\square$

Example 31.7 (Catalan asymptotics). The Catalan generating function is

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z} = \frac{2}{1 + \sqrt{1 - 4z}} = 2 - 2(1 - 4z)^{1/2} + O(1 - 4z).$$

The nonanalytic term has $\rho = 1/4$, $\alpha = -1/2$, and multiplier -2 . Since $\Gamma(-1/2) = -2\sqrt{\pi}$,

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}. \quad (31.13)$$

This proof also explains the universal $n^{-3/2}$ factor produced by a simple square-root branch point.

Part IX

Pochhammer, Polygamma, and Barnes Calculus

Chapter 32

Pochhammer symbols and parameter differentiation

32.1 Rising and falling factorials

For $n \geq 0$, the rising Pochhammer symbol is

$$(a)_n = a(a+1) \cdots (a+n-1), \quad (a)_0 = 1. \quad (32.1)$$

It is related to the factorial powers used earlier by

$$(a)_n = a^{\overline{n}} = \frac{\Gamma(a+n)}{\Gamma(a)}, \quad x^{\overline{n}} = (-1)^n (-x)_n. \quad (32.2)$$

The gamma quotient is initially valid away from poles and extends polynomially in a for fixed n . Expanding in powers of a gives

$$(a)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} a^k, \quad (32.3)$$

so unsigned first-kind Stirling numbers are precisely the monomial coefficients of the Pochhammer polynomial.

32.2 The binomial theorem and Vandermonde convolution

Theorem 32.1 (Generalized binomial series). *For formal z , or analytically for $|z| < 1$,*

$$(1-z)^{-a} = \sum_{n \geq 0} \frac{(a)_n}{n!} z^n. \quad (32.4)$$

Proof. Let $F(z)$ be the series on the right. Since $(a)_{n+1} = (a+n)(a)_n$, coefficient comparison gives

$$(1-z)F'(z) = aF(z), \quad F(0) = 1.$$

The unique formal solution of this differential equation is $(1-z)^{-a}$: logarithmic differentiation gives $F'/F = a/(1-z)$ and the initial value fixes the multiplicative constant. For $|z| < 1$, the formal solution is the analytic branch with value 1 at zero. $\square$

Corollary 32.2 (Vandermonde convolution). *For $n \geq 0$,*

$$(a+b)_n = \sum_{k=0}^n \binom{n}{k} (a)_k (b)_{n-k}. \quad (32.5)$$

Proof. Multiply the two binomial series for $(1 - z)^{-a}$ and $(1 - z)^{-b}$. Their product is $(1 - z)^{-(a+b)}$; comparison of exponential-normalized coefficients gives the result. $\square$

Proposition 32.3 (Beta integral). *If $\Re a > 0$ and $n \geq 0$, then*

$$\int_0^1 t^{a-1} (1-t)^n dt = \frac{n!}{(a)_{n+1}}. \quad (32.6)$$

Proof. The integral is $B(a, n+1) = \Gamma(a)\Gamma(n+1)/\Gamma(a+n+1)$. Use $\Gamma(n+1) = n!$ and $(a)_{n+1} = \Gamma(a+n+1)/\Gamma(a)$. $\square$

32.3 Parameter derivatives and complete Bell polynomials

Let $\psi(z) = \Gamma'(z)/\Gamma(z)$ and let $\psi^{(r)}$ be its r th derivative. At points where the two terms are finite, define

$$\Delta_r(a, n) = \psi^{(r)}(a+n) - \psi^{(r)}(a). \quad (32.7)$$

For an integer $n \geq 1$, the finite-sum formula below supplies its meromorphic continuation when the separate polygamma values have poles.

Theorem 32.4 (All parameter derivatives of a Pochhammer symbol). *For $n \geq 0$, $m \geq 1$, and $a \notin \{0, -1, \dots, -n+1\}$,*

$$\frac{\partial^m}{\partial a^m} (a)_n = (a)_n \mathcal{B}_m(\Delta_0(a, n), \Delta_1(a, n), \dots, \Delta_{m-1}(a, n)). \quad (32.8)$$

The right side has removable singularities at the excluded points; its unique polynomial continuation there equals the derivative on the left. The case $m = 0$ is the identity $\partial_a^0 (a)_n = (a)_n$. For positive integer n ,

$$\Delta_r(a, n) = (-1)^r r! \sum_{j=0}^{n-1} \frac{1}{(a+j)^{r+1}}. \quad (32.9)$$

Proof. Let $L(a) = \log(a)_n = \log \Gamma(a+n) - \log \Gamma(a)$. Then $L^{(j)}(a) = \Delta_{j-1}(a, n)$ for $j \geq 1$. Since $(a)_n = e^{L(a)}$, Faà di Bruno's formula for the exponential outer function gives (32.8). The gamma recurrence implies $\psi(z+1) - \psi(z) = 1/z$. Differentiate r times and telescope from a to $a+n$ to obtain (32.9). $\square$

The first three cases are

$$\partial_a (a)_n = (a)_n \Delta_0, \quad (32.10)$$

$$\partial_a^2 (a)_n = (a)_n (\Delta_0^2 + \Delta_1), \quad (32.11)$$

$$\partial_a^3 (a)_n = (a)_n (\Delta_0^3 + 3\Delta_0\Delta_1 + \Delta_2). \quad (32.12)$$

Thus Bell polynomials organize derivatives with respect to a hypergeometric parameter exactly as they organize derivatives of ordinary composites.

Corollary 32.5 (Derivatives of reciprocal Pochhammer symbols). *For $m \geq 1$ and $(a)_n \neq 0$,*

$$\frac{\partial^m}{\partial a^m} \frac{1}{(a)_n} = \frac{1}{(a)_n} \mathcal{B}_m(-\Delta_0, -\Delta_1, \dots, -\Delta_{m-1}). \quad (32.13)$$

For $m = 0$ both sides are simply $(a)_n^{-1}$.

Proof. Write $(a)_n^{-1} = e^{-L(a)}$ and apply the same exponential form of Faà di Bruno's formula. $\square$

Chapter 33

Digamma and polygamma functions from first principles

33.1 Euler's limiting product for the gamma function

Theorem 33.1 (Euler limit). For $\Re z > 0$,

$$\Gamma(z) = \lim_{N \rightarrow \infty} \frac{N! N^z}{z(z+1) \cdots (z+N)}, \quad (33.1)$$

locally uniformly in z on the right half-plane.

Proof. The beta integral gives

$$B(z, N+1) = \int_0^1 t^{z-1} (1-t)^N dt = \frac{N!}{z(z+1) \cdots (z+N)}.$$

After multiplying by N^z and substituting $u = Nt$,

$$N^z B(z, N+1) = \int_0^N u^{z-1} \left(1 - \frac{u}{N}\right)^N du.$$

For $0 \leq u \leq N$, $(1 - u/N)^N \leq e^{-u}$. On compact subsets of $\Re z > 0$, the integrand is dominated by an integrable multiple of $u^{\sigma-1} e^{-u}$ near zero and infinity. It converges pointwise to $u^{z-1} e^{-u}$, so dominated convergence, uniformly on compact subsets, gives the gamma integral. $\square$

33.2 Series for the logarithmic derivatives

Let γ be Euler's constant.

Theorem 33.2 (Digamma and polygamma series). For $z \notin \{0, -1, -2, \dots\}$,

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+z} \right). \quad (33.2)$$

For $m \geq 1$,

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} \frac{1}{(k+z)^{m+1}}. \quad (33.3)$$

Both series converge locally uniformly away from the poles.

Proof. Take the logarithmic derivative of the finite expression in (33.1):

$$\frac{d}{dz} \log \left(\frac{N! N^z}{z(z+1) \cdots (z+N)} \right) = \log N - \sum_{k=0}^N \frac{1}{k+z}.$$

Rewrite the right side as

$$(\log N - H_{N+1}) + \sum_{k=0}^N \left(\frac{1}{k+1} - \frac{1}{k+z} \right).$$

The first term tends to $-\gamma$. On compact subsets of the right half-plane, the Euler approximants converge locally uniformly to the nonvanishing function Γ ; Cauchy's integral formula therefore gives convergence of their logarithmic derivatives. The summand in the second term is $O(k^{-2})$ locally uniformly in z , so the limit may be summed termwise. This proves (33.2) first for $\Re z > 0$. The series on the right converges locally uniformly on $\mathbb{C} \setminus \{0, -1, -2, \dots\}$ and defines a meromorphic function there; analytic continuation gives the asserted identity throughout that domain. Termwise differentiation, justified locally uniformly away from the poles, gives (33.3). $\square$

Corollary 33.3 (Recurrences and integer values). *For $m \geq 0$,*

$$\psi(z+1) = \psi(z) + \frac{1}{z}, \quad (33.4)$$

$$\psi^{(m)}(z+1) = \psi^{(m)}(z) + (-1)^m \frac{m!}{z^{m+1}}. \quad (33.5)$$

For $n \geq 0$,

$$\psi(n+1) = H_n - \gamma, \quad (33.6)$$

$$\psi^{(m)}(n+1) = (-1)^{m+1} m! \left(\zeta(m+1) - H_n^{(m+1)} \right), \quad m \geq 1. \quad (33.7)$$

Proof. Subtract the series at z from the series at $z+1$; all terms telescope. Starting from $\psi(1) = -\gamma$ and $\psi^{(m)}(1) = (-1)^{m+1} m! \zeta(m+1)$ and iterating the recurrences gives the integer values. $\square$

33.3 Reflection and multiplication

Theorem 33.4 (Reflection). *For $z \notin \mathbb{Z}$,*

$$\psi(1-z) - \psi(z) = \pi \cot(\pi z). \quad (33.8)$$

More generally, for $m \geq 0$,

$$(-1)^m \psi^{(m)}(1-z) - \psi^{(m)}(z) = \frac{d^m}{dz^m} (\pi \cot(\pi z)). \quad (33.9)$$

Proof. Subtract the two digamma series:

$$\psi(1-z) - \psi(z) = \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{k+z} - \frac{1}{k-z} \right).$$

The Weierstrass product $\sin(\pi z) = \pi z \prod_{k \geq 1} (1 - z^2/k^2)$ has locally uniform logarithmic derivative

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{k+z} - \frac{1}{k-z} \right),$$

which proves the first formula. Repeated differentiation gives the second. $\square$

Theorem 33.5 (Multiplication). *For a positive integer q ,*

$$\sum_{r=0}^{q-1} \psi\left(z + \frac{r}{q}\right) = q\psi(qz) - q \log q, \quad (33.10)$$

$$\sum_{r=0}^{q-1} \psi^{(m)}\left(z + \frac{r}{q}\right) = q^{m+1} \psi^{(m)}(qz), \quad m \geq 1. \quad (33.11)$$

Proof. For $m \geq 1$, substitute (33.3) and write every nonnegative integer uniquely as $qk + r$:

$$\sum_{r=0}^{q-1} \sum_{k \geq 0} \left(z + \frac{r}{q} + k\right)^{-m-1} = q^{m+1} \sum_{j \geq 0} (qz + j)^{-m-1}.$$

This proves (33.11). The derivative with respect to z of the difference between the two sides of (33.10) is zero by the case $m = 1$, so that difference is constant. As $z \rightarrow +\infty$, the Euler–Maclaurin estimate $\psi(z) = \log z + O(1/z)$ shows that the constant is $-q \log q$. $\square$

For $x > 0$, (33.3) also gives the sign law

$$(-1)^{m+1} \psi^{(m)}(x) > 0 \quad (m \geq 1), \quad (33.12)$$

which is frequently useful when proving convexity of gamma and Pochhammer ratios.

Chapter 34

Barnes' G -function and superfactorial asymptotics

34.1 Canonical product and recurrence

Barnes' G -function is the entire function defined by the canonical product of Barnes [14]; the normalization and asymptotic conventions below agree with [13].

$$G(z+1) = (2\pi)^{z/2} \exp\left(-\frac{z + (1+\gamma)z^2}{2}\right) \times \prod_{k=1}^{\infty} \left\{ \left(1 + \frac{z}{k}\right)^k \exp\left(\frac{z^2}{2k} - z\right) \right\}. \quad (34.1)$$

The logarithm of the k th factor is $O(k^{-2})$ uniformly on compact z -sets, so the product converges locally uniformly and defines an entire function.

Theorem 34.1 (Barnes recurrence). *The product (34.1) satisfies*

$$G(z+1) = \Gamma(z)G(z), \quad G(1) = 1. \quad (34.2)$$

Consequently, for integers $n \geq 1$,

$$G(n+1) = \prod_{k=1}^{n-1} k! = \prod_{j=1}^{n-1} j^{n-j}. \quad (34.3)$$

Proof. Let $P(z)$ denote the right side of (34.1), so that $P(z) = G(z+1)$, and let P_N be the product truncated after $k = N$. Setting $z = 0$ gives $P(0) = 1$. Direct division of the two truncated products gives

$$\frac{P_N(z)}{P_N(z-1)} = \sqrt{2\pi} \exp\left(-(1+\gamma)z + \frac{\gamma}{2}\right) \prod_{k=1}^N \left(\frac{k+z}{k+z-1}\right)^k \exp\left(\frac{2z-1}{2k} - 1\right).$$

The algebraic product telescopes:

$$\prod_{k=1}^N \left(\frac{k+z}{k+z-1}\right)^k = \frac{(N+z)^N}{z \prod_{k=1}^{N-1} (k+z)} = \Gamma(z) \frac{(N+z)^N}{\Gamma(N+z)}.$$

Hence

$$\frac{P_N(z)}{P_N(z-1)} = \Gamma(z) A_N(z), \quad (34.4)$$

where

$$A_N(z) = \sqrt{2\pi} e^{-(1+\gamma)z + \gamma/2} \frac{(N+z)^N}{\Gamma(N+z)} e^{(z-1/2)H_N - N}.$$

The classical estimates $H_N = \log N + \gamma + O(N^{-1})$ and

$$\Gamma(N+z) = \sqrt{2\pi} (N+z)^{N+z-1/2} e^{-(N+z)} (1 + O(N^{-1}))$$

hold locally uniformly for bounded z . Substitution shows

$$A_N(z) = \left(\frac{N}{N+z} \right)^{z-1/2} (1 + O(N^{-1})) \rightarrow 1;$$

all constant exponential factors cancel exactly. Local uniform convergence of the canonical products now lets $N \rightarrow \infty$ in (34.4), yielding $P(z)/P(z-1) = \Gamma(z)$ away from the poles; analytic continuation gives the functional equation everywhere it is defined. Since $P(0) = 1$, this is $G(z+1) = \Gamma(z)G(z)$ with $G(1) = 1$.

Iteration gives $G(n+1) = \Gamma(n) \cdots \Gamma(1) = \prod_{k=1}^{n-1} k!$. A fixed integer j occurs in $j!$, $(j+1)!$, $\dots$, $(n-1)!$, exactly $n-j$ times, giving the second product in (34.3). $\square$

Termwise logarithmic differentiation of the canonical product gives the convergent representation

$$\frac{d}{dz} \log G(z+1) = \frac{1}{2} \log(2\pi) - \frac{1}{2} - (1+\gamma)z + \sum_{k=1}^{\infty} \left(\frac{k}{k+z} - 1 + \frac{z}{k} \right). \quad (34.5)$$

It is the order-two analogue of the digamma series.

34.2 Asymptotic expansion

Theorem 34.2 (Barnes asymptotic expansion on the positive integers). *For each fixed $p \geq 1$, as $N \rightarrow \infty$ through positive integers,*

$$\begin{aligned} \log G(N+1) &= \frac{N^2}{2} \log N - \frac{3N^2}{4} + \frac{N}{2} \log(2\pi) - \frac{1}{12} \log N + \zeta'(-1) \\ &\quad + \sum_{r=1}^{p-1} \frac{\beta_{2r+2}}{2r(2r+2)N^{2r}} + O(N^{-2p}). \end{aligned} \quad (34.6)$$

Equivalently, if A is the Glaisher–Kinkelin constant, then

$$\log A = \frac{1}{12} - \zeta'(-1). \quad (34.7)$$

Proof. From (34.3),

$$\log G(N+1) = N \log \Gamma(N) - \sum_{k=1}^{N-1} k \log k. \quad (34.8)$$

Apply Euler–Maclaurin to $\log x$ and to $x \log x$. The first application gives Stirling's expansion

$$\log \Gamma(N) = \left(N - \frac{1}{2} \right) \log N - N + \frac{1}{2} \log(2\pi) + \sum_{r=1}^p \frac{\beta_{2r}}{2r(2r-1)N^{2r-1}} + O(N^{-2p-1}). \quad (34.9)$$

For the second sum, apply Euler–Maclaurin to the tail of $\zeta(s) = \sum_{k \geq 1} k^{-s}$, initially for $\Re s > 1$ and then by analytic continuation. Uniformly for s near -1 ,

$$\begin{aligned} \sum_{k=1}^{N-1} k^{-s} &= \zeta(s) - \frac{N^{1-s}}{s-1} - \frac{1}{2}N^{-s} \\ &\quad - \sum_{r=1}^{p+1} \frac{\beta_{2r}}{(2r)!} (s)_{2r-1} N^{1-s-2r} + O(N^{-\Re s - 2p-1}), \end{aligned}$$

with the same bound after one s -derivative (up to a harmless logarithm that is absorbed after retaining the next term). Since

$$-\frac{\partial}{\partial s} \sum_{k=1}^{N-1} k^{-s} \Big|_{s=-1} = \sum_{k=1}^{N-1} k \log k,$$

differentiate the displayed continuation at $s = -1$. The first three tail terms contribute, respectively,

$$\frac{N^2}{2} \log N - \frac{N^2}{4}, \quad -\frac{N}{2} \log N, \quad \frac{1}{12} \log N + \frac{1}{12}.$$

For $r \geq 2$, $(s)_{2r-1}$ contains the single factor $s+1$ and

$$\frac{d}{ds} (s)_{2r-1} \Big|_{s=-1} = -(2r-3)!.$$

Reindexing $r \mapsto r+1$ therefore gives

$$\begin{aligned} \sum_{k=1}^{N-1} k \log k &= \frac{N^2}{2} \log N - \frac{N^2}{4} - \frac{N}{2} \log N + \frac{1}{12} \log N \\ &\quad + \frac{1}{12} - \zeta'(-1) + \sum_{r=1}^{p-1} d_r N^{-2r} + O(N^{-2p}), \end{aligned} \tag{34.10}$$

where

$$d_r = -\frac{\beta_{2r+2}}{2r(2r+1)(2r+2)}.$$

Substitute (34.9) and (34.10) into (34.8). The $N \log N$ terms cancel, the constants $1/12$ cancel, and the two Bernoulli contributions combine to $\beta_{2r+2}/(2r(2r+2))$. This yields (34.6). The Euler–Maclaurin remainder bound from Theorem 31.4 gives the stated error after the two applications.

The traditional constant A is defined by the hyperfactorial limit

$$A = \lim_{N \rightarrow \infty} \frac{\prod_{k=1}^{N-1} k^k}{N^{N^2/2 - N/2 + 1/12} e^{-N^2/4}}.$$

Taking logarithms in (34.10) identifies its constant as $1/12 - \zeta'(-1)$, proving (34.7). $\square$

The first correction is

$$\frac{\beta_4}{2 \cdot 4N^2} = -\frac{1}{240N^2},$$

so the expansion begins

$$\begin{aligned} \log G(N+1) &= \frac{N^2}{2} \log N - \frac{3N^2}{4} + \frac{N}{2} \log(2\pi) - \frac{1}{12} \log N + \zeta'(-1) \\ &\quad - \frac{1}{240N^2} + \frac{1}{1008N^4} + O(N^{-6}). \end{aligned}$$

The Bernoulli coefficients therefore link Barnes' superfactorial growth directly to the same summation calculus that controls Stirling's expansion for Γ .

Part X

Extended Inversion and Analytic Applications

Chapter 35

Inverse functions beyond the basic examples

35.1 Inverse trigonometric and hyperbolic series

Theorem 35.1 (Four classical inverse series). For $|y| < 1$,

$$\arcsin y = \sum_{m \geq 0} \frac{1}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}, \quad (35.1)$$

$$\arctan y = \sum_{m \geq 0} \frac{(-1)^m}{2m+1} y^{2m+1}, \quad (35.2)$$

$$\operatorname{arsinh} y = \sum_{m \geq 0} \frac{(-1)^m}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}, \quad (35.3)$$

$$\operatorname{artanh} y = \sum_{m \geq 0} \frac{y^{2m+1}}{2m+1}. \quad (35.4)$$

Consequently, Lagrange inversion gives the coefficient evaluations

$$[x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1} = \frac{1}{4^m} \binom{2m}{m}, \quad (35.5)$$

$$[x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1} = (-1)^m, \quad (35.6)$$

$$[x^{2m}] \left(\frac{x}{\sinh x} \right)^{2m+1} = (-1)^m \frac{1}{4^m} \binom{2m}{m}, \quad (35.7)$$

$$[x^{2m}] \left(\frac{x}{\tanh x} \right)^{2m+1} = 1. \quad (35.8)$$

Proof. The derivatives of the four inverse functions are, respectively,

$$(1-y^2)^{-1/2}, \quad (1+y^2)^{-1}, \quad (1+y^2)^{-1/2}, \quad (1-y^2)^{-1}.$$

Expand them by the generalized binomial theorem or the geometric series and integrate term by term from 0 to y . This proves (35.1)–(35.4).

If $g = f^{(-1)}$ with $f(0) = 0$ and $f'(0) = 1$, Lagrange inversion says

$$[y^n]g(y) = \frac{1}{n} [x^{n-1}] \left(\frac{x}{f(x)} \right)^n.$$

Apply this with $f = \sin, \tan, \sinh, \tanh$, take $n = 2m+1$, and compare with the four explicit inverse coefficients. $\square$

35.2 Diagonal identities for generalized Bernoulli and Cauchy series

Proposition 35.2 (Exponential–logarithmic diagonal identities). *For $n \geq 1$,*

$$[x^{n-1}] \left(\frac{x}{e^x - 1} \right)^n = (-1)^{n-1}, \quad (35.9)$$

$$[x^{n-1}] \left(\frac{x}{\log(1+x)} \right)^n = \frac{1}{(n-1)!}. \quad (35.10)$$

Proof. The inverse of $e^x - 1$ is $\log(1+y)$, whose coefficient of y^n is $(-1)^{n-1}/n$. Lagrange inversion therefore gives

$$\frac{1}{n} [x^{n-1}] \left(\frac{x}{e^x - 1} \right)^n = \frac{(-1)^{n-1}}{n}.$$

This proves the first identity. Conversely, the inverse of $\log(1+x)$ is $e^y - 1$, whose coefficient of y^n is $1/n!$. A second application gives the second identity. $\square$

35.3 Implicit equations with an offset

Theorem 35.3 (Offset Lagrange–Bürmann formula). *Suppose $u(t)$ is determined by*

$$u = a + t\Phi(u), \quad \Phi(a) \neq 0. \quad (35.11)$$

Then for $n \geq 1$ and any formal or analytic H ,

$$\boxed{[t^n]H(u(t)) = \frac{1}{n}[z^{n-1}]H'(a+z)\Phi(a+z)^n.} \quad (35.12)$$

Proof. Put $w = u - a$ and $\phi(w) = \Phi(a+w)$. Then $w = t\phi(w)$ and $H(u) = \tilde{H}(w)$ with $\tilde{H}(w) = H(a+w)$. The ordinary Lagrange–Bürmann formula gives

$$[t^n]\tilde{H}(w(t)) = \frac{1}{n}[z^{n-1}]\tilde{H}'(z)\phi(z)^n,$$

which is exactly (35.12). $\square$

Chapter 36

Narayana refinement of the Catalan family

The Catalan material in the supplemental Wikipedia files admits a useful bivariate refinement. Let $N(n, k)$ count Dyck paths of semilength n with exactly k peaks.

Theorem 36.1 (Narayana numbers). *For $1 \leq k \leq n$,*

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}. \quad (36.1)$$

Thus $\sum_{k=1}^n N(n, k) = C_n$.

Proof. Let $C(z, u)$ be the generating function in which z marks semilength and u marks peaks, including the empty path. A nonempty Dyck path decomposes uniquely as $UADB$, where A and B are Dyck paths. If A is empty, the initial UD is a new peak; if A is nonempty, it is not. Therefore

$$C = 1 + zC(C + u - 1). \quad (36.2)$$

Put $W = C - 1$. Then $W = z(1 + W)(u + W)$. Lagrange inversion gives

$$\begin{aligned} [z^n u^k] W &= \frac{1}{n} [w^{n-1} u^k] (1 + w)^n (u + w)^n \\ &= \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \end{aligned}$$

which proves (36.1). Setting $u = 1$ makes (36.2) the Catalan equation $C = 1 + zC^2$, so the row sum is C_n . $\square$

The Narayana polynomial $\sum_k N(n, k)u^k$ is symmetric because $N(n, k) = N(n, n+1-k)$. In the standard geometric interpretation, the same row is the h -vector of the associahedron, linking the bivariate Catalan refinement to the face structure already used in associahedral inversion.

Chapter 37

Good's multivariate inversion formula

Let $\mathbf{w} = (w_1, \dots, w_d)$ be the unique formal solution of

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0, \quad 1 \leq i \leq d. \quad (37.1)$$

For a multi-index $\mathbf{n} = (n_1, \dots, n_d)$, write $\mathbf{t}^{\mathbf{n}} = \prod_i t_i^{n_i}$ and $\phi^{\mathbf{n}} = \prod_i \phi_i^{n_i}$.

Theorem 37.1 (Good's formula [16]). *For every formal series F and every $\mathbf{n} \in \mathbb{N}^d$,*

$$[\mathbf{t}^{\mathbf{n}}]F(\mathbf{w}(\mathbf{t})) = [\mathbf{x}^{\mathbf{n}}]F(\mathbf{x})\phi(\mathbf{x})^{\mathbf{n}} \times \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (37.2)$$

Proof. Represent the coefficient on the left by an iterated formal residue:

$$\text{Res}_{\mathbf{t}} F(\mathbf{w}(\mathbf{t})) \prod_{i=1}^d t_i^{-n_i-1} dt_1 \cdots dt_d.$$

Make the formal change of variables $t_i = x_i / \phi_i(\mathbf{x})$. Its Jacobian has entries

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

Because (37.1) implies $\mathbf{w}(\mathbf{t}(\mathbf{x})) = \mathbf{x}$, the transformed residue is

$$\text{Res}_{\mathbf{x}} F(\mathbf{x}) \prod_i \phi_i(\mathbf{x})^{n_i} x_i^{-n_i-1} \det \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right) d\mathbf{x}.$$

Let $X = \text{diag}(x_1, \dots, x_d)$ and $Y = \text{diag}(\phi_1^{-1}, \dots, \phi_d^{-1})J_{\phi}$. The determinant displayed here is $\det(I - XY)$. The identity $\det(I - XY) = \det(I - YX)$ changes it into the determinant in (37.2), whose (i, j) entry contains x_j . Reading the final residue as a coefficient proves the theorem. $\square$

Remark 37.2. For $d = 1$ the determinant is $1 - x\phi'(x)/\phi(x)$. Formal integration by parts reduces Good's formula to the one-variable Lagrange–Bürmann theorem. The determinant is therefore the genuine multivariate Jacobian correction, not an optional decoration.

Chapter 38

Kepler's equation and Fourier–Bessel reversion

For eccentricity $0 \leq e < 1$, Kepler's equation is

$$E = M + e \sin E. \quad (38.1)$$

The shifted reversion theorem immediately gives

$$E = M + \sum_{k \geq 1} \frac{e^k}{k!} \left(\frac{d}{dM} \right)^{k-1} \sin^k M. \quad (38.2)$$

The same solution has a global Fourier expansion.

Theorem 38.1 (Fourier–Bessel solution of Kepler's equation). *For $0 \leq e < 1$,*

$$E = M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM), \quad (38.3)$$

where J_n is the Bessel function of the first kind.

Proof. The map $E \mapsto M = E - e \sin E$ has derivative $1 - e \cos E > 0$, so it is strictly increasing and its inverse satisfies $E(M + 2\pi) = E(M) + 2\pi$. Thus $q(M) = E(M) - M$ is a smooth 2π -periodic odd function. Let c_n be its sine coefficient. Since $q = e \sin E$ and $dM = (1 - e \cos E) dE$,

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^{2\pi} q(M) \sin(nM) dM \\ &= \frac{e}{\pi} \int_0^{2\pi} \sin E \sin(n(E - e \sin E)) (1 - e \cos E) dE. \end{aligned}$$

Now

$$\frac{d}{dE} \cos(n(E - e \sin E)) = -n(1 - e \cos E) \sin(n(E - e \sin E)).$$

Integration by parts, with zero boundary term, gives

$$c_n = \frac{e}{\pi n} \int_0^{2\pi} \cos E \cos(nE - ne \sin E) dE.$$

Using $2 \cos E \cos A = \cos(A + E) + \cos(A - E)$ and the integral representation

$$J_m(t) = \frac{1}{2\pi} \int_0^{2\pi} \cos(mE - t \sin E) dE,$$

we obtain

$$c_n = \frac{e}{n} (J_{n-1}(ne) + J_{n+1}(ne)).$$

The Bessel recurrence $J_{n-1}(t) + J_{n+1}(t) = 2nJ_n(t)/t$ gives $c_n = 2J_n(ne)/n$. The smoothness of q justifies Fourier reconstruction and proves (38.3). $\square$

Chapter 39

Critical values and square-root inverse singularities

Theorem 39.1 (Local square-root form). *Let f be analytic near a , with*

$$f(a) = b, \quad f'(a) = 0, \quad f''(a) \neq 0.$$

Then the two local inverse branches near b have convergent Puiseux expansions

$$z = a + \sum_{m \geq 1} c_m (w - b)^{m/2}, \quad (39.1)$$

with opposite signs for c_1 , and

$$c_1 = \pm \sqrt{\frac{2}{f''(a)}}. \quad (39.2)$$

Proof. Write $z = a + u$. Taylor's theorem factors

$$f(a + u) - b = u^2 h(u), \quad h(0) = \frac{1}{2} f''(a) \neq 0,$$

with h analytic. Choose an analytic square root of h near zero and put $v = u\sqrt{h(u)}$. Then $v'(0) = \sqrt{h(0)} \neq 0$, so the analytic inverse function theorem gives an analytic inverse $u = \chi(v)$. Since $v^2 = w - b$, the two choices $v = \pm\sqrt{w - b}$ give $z = a + \chi(\pm\sqrt{w - b})$, a convergent series in half-integer powers. The linear term is $u = v/\sqrt{h(0)} + O(v^2)$, proving (39.2). $\square$

Corollary 39.2 (Coefficient signature of a simple critical point). *Suppose one branch in (39.1) has its unique dominant singularity at $w = \rho$, locally contains a nonzero term $\lambda(1 - w/\rho)^{1/2}$, and its remaining local expansion satisfies the analytic-multiplier hypotheses of Theorem 31.6. Then its Taylor coefficients have leading form*

$$[w^n]z(w) \sim -\frac{\lambda}{2\sqrt{\pi}} \rho^{-n} n^{-3/2}. \quad (39.3)$$

Proof. Apply the Darboux transfer theorem with $\alpha = -1/2$ and use $\Gamma(-1/2) = -2\sqrt{\pi}$. $\square$

This principle explains several radii and coefficient exponents at once. For $f(z) = ze^z$, the critical point $z = -1$ maps to $-e^{-1}$ and produces the square-root branch of Lambert W . The Catalan solution has its quadratic critical value at $1/4$. The inverse sine meets critical points of $\sin z$ at $\pm\pi/2$, whose images are ± 1 . Formal inversion determines all coefficients without convergence assumptions; analytic continuation and the nearest singular or critical value determine their eventual size.

Chapter 40

Online research update and connections

The online search used for this edition was restricted, for technical claims, to primary or authoritative sources. The current NIST Digital Library of Mathematical Functions chapters on Bernoulli and Euler polynomials, gamma and polygamma functions, and Barnes' G -function supplied a normalization check for the special-function chapters [13]. Gessel's survey of Lagrange inversion provided a modern cross-check on the formal-residue and multivariate forms [15], while Good's original paper was used for the determinant normalization in [Theorem 37.1](#) [16]. A 2026 paper by Hwang, Li, and Zacharovas develops elementary local and central limit theorems for second-kind Stirling numbers and effective Bell-number asymptotics [17]. Its results are compatible with the saddle-point and real-rootedness framework proved earlier; it is recorded as a recent continuation rather than silently folded into older attribution.

The additions in this part show three broader links that were not fully articulated in any one draft:

1. inverse-function coefficients, Narayana refinements, and Abel polynomials are all direct Lagrange extractions;
2. a simple critical point of the forward map forces a square-root singularity of the inverse and hence an $n^{-3/2}$ coefficient factor;
3. Good's determinant is the multivariate Jacobian counterpart of the one-variable integration-by-parts term.

Part XI

Source Merge, Deduplication, and Reference Record

Chapter 41

Five-draft merge and deduplication ledger

The supplied archive contains five substantial LaTeX drafts rather than five independent topics. Each draft covers the same central families—Stirling, Bell, Eulerian, Faà di Bruno, and Lagrange formulae—but emphasizes different proofs, applications, or normalizations. A literal concatenation would repeat most definitions and master theorems several times. The present article uses a theorem-level merge: one canonical statement and proof is retained for each result, while genuinely distinct applications, algorithms, examples, and proof methods are integrated at the point where they become mathematically natural.

Draft	Principal contribution retained	Editorial treatment
<code>combinatorial_coefficient_calculus.tex</code>	The broadest proof-oriented development: formal series, both Stirling triangles, Bell numbers and polynomials, Eulerian families, composition, reversion, congruences, asymptotics, and archive corrections.	Used as the structural spine. Repeated material from the other drafts was compared against it theorem by theorem; stronger hypotheses, clearer proofs, or distinct applications were imported, while duplicate introductions and tables were removed.
<code>Combinatorial_Formulae_and_Inversion_Theorems.tex</code>	Binomial inversion, a clean set-partition and permutation foundation, moment/factorial-moment language, reverse Stirling recurrences, integral transforms, and explicit special-function series.	Binomial and partition-lattice inversion were expanded in Chapter 25. Probability and special-function material was merged into the existing Stirling and Bell chapters rather than repeated as a second survey.

Continued on the next page

Draft	Principal contribution retained	Editorial treatment
Formulae_Compensum.tex	Inverse trigonometric examples, arbitrary-point and offset inversion, Good’s multivariate theorem, partition-lattice Möbius inversion, binomial-type and Sheffer sequences, exponential Riordan arrays, Abel polynomials, and triangular coefficient algorithms.	These were the largest genuinely nonduplicated block. They now form Parts VII and X; cross-references that depended on the compendium’s private numbering were replaced by self-contained statements and proofs.
Formulae_Monograph.tex in Formulae_Monograph_Package	Notation tables, normalization warnings, finite computational certificates, low-order composition and inversion checks, and a concise dependency map.	The normalization conventions were adopted globally. Independent inverse coefficients through degree five are proved in Theorem 27.3; software-specific lists were replaced by implementation-neutral recurrences.
Formulae_Monograph.tex in Formulae_Monograph_Source_and_PDF	The most detailed distributional decoding of the delta-function reversion argument, several-variable differentiation, analytic inverse diagnostics, and the Fourier–Bessel solution of Kepler’s equation.	The rigorous change-of-variables interpretation of the delta proof was retained in the core reversion chapter. Kepler’s equation, absent from the spine draft, is proved in Chapter 38; duplicated basic inversion theory was removed.

41.1 Deduplication rules

The merge followed five explicit rules.

1. *One definition per object.* Signed and unsigned first-kind Stirling numbers, ordinary and exponential Bell polynomials, Bell versus Bernoulli notation, and type- B Eulerian indexing are fixed once and used throughout.
2. *One master proof, many corollaries.* The exponential formula, finite differences, Faà di Bruno, formal residues, and Lagrange–Bürmann inversion are proved once. Repeated formulas are then stated as specializations rather than re-proved under new symbols.
3. *Distinct proof methods are retained only when they add information.* For example, Kepler’s equation keeps both the local reversion series and the global Fourier–Bessel resummation; Bell-number asymptotics keeps elementary bounds and saddle-point analysis because they answer different questions.
4. *Computational claims receive certificates.* Finite tables, modular searches, and low-order inverse coefficients are accompanied by exact recurrences or direct substitution checks.
5. *Schematic statements are repaired, not silently copied.* Missing ranges, index shifts, suppressed

Jacobians, undefined asymptotic coefficients, and formal-versus-analytic ambiguities are made explicit.

Chapter 42

Supplemental directories and newly integrated material

42.1 Wikipedia directory

The plain-text Wikipedia captures were treated as topic maps rather than as text to reproduce. Their mathematical material is integrated as follows.

Supplemental topic	Integration
Stirling numbers, Bell numbers, Bell polynomials, Eulerian numbers	Their standard interpretations, generating functions, recurrences, and normalization variants were checked against the core chapters. Repeated encyclopedia lists were consolidated into theorem families with proofs.
Faà di Bruno's formula	Set-partition, multiplicity, Bell-polynomial, exponential-series, and multivariate forms appear in the differentiation and composition part.
Lagrange inversion and reversion	Formal residue, analytic inverse, shifted, denominator, Bell-polynomial, and multivariate forms are separated by hypotheses and normalization.
Catalan numbers and associahedra	Catalan and Fuss–Catalan inversion, plane trees, associahedral alternating face sums, and the new Narayana peak refinement are developed in the inversion chapters.

42.2 MathWorld directory

The MathWorld PDFs were accompanied by Mathematica notebook source, allowing formula-level inspection without depending on raster extraction. The material added beyond the five drafts includes:

1. Bernoulli polynomials, the positive- B_1 convention, the hyperbolic-sine modified Bernoulli logarithm, and Bernoulli polynomials of the second kind;
2. Euler polynomials, Genocchi numbers and polynomials, and alternating power sums;
3. complementary Bell numbers, their Dobiński series, recurrence, and reciprocal Bell transform;
4. Darboux's weighted Taylor formula, Euler–Maclaurin summation with an explicit periodic-Bernoulli remainder, and the distinct Darboux coefficient-transfer theorem;
5. Pochhammer symbols, all parameter derivatives through Bell polynomials, digamma and polygamma series, and Barnes' G -function with its superfactorial asymptotic expansion.

All of these items are derived in Parts VIII and IX; notebook output was used as a discovery aid, not as a substitute for proof.

42.3 Additional online research

Current online verification used authoritative or primary sources. The NIST Digital Library of Mathematical Functions supplied current normalization and asymptotic checks for Bernoulli and Euler polynomials, polygamma functions, and Barnes' G -function [13]. Gessel's survey supplied a modern reference for the formal and combinatorial forms of Lagrange inversion [15], and Good's original article fixed the multivariate determinant normalization [16]. The 2026 work of Hwang, Li, and Zacharovas was added to the asymptotic bibliography as a recent extension of central-range Stirling analysis [17].

Chapter 43

Correction and proof-completeness record

43.1 Normalization corrections

The following source-level ambiguities are resolved globally.

1. $s(n, k)$ is signed and $\begin{bmatrix} n \\ k \end{bmatrix}$ is unsigned. They are never interchanged inside a logarithmic generating function or a factorial expansion.
2. Bell numbers are B_n , partial and complete exponential Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$, Bernoulli numbers are β_n , Euler polynomials are $\mathcal{E}_n(x)$, and complementary Bell numbers are C_n .
3. Type- B Eulerian numbers count descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This convention makes the corrected explicit formula agree with the recurrence and printed rows.
4. Ordinary and exponential inverse coefficients are separated by their factorial normalization; the associahedral face formula is stated for an ordinary series.
5. Good's determinant includes the full Jacobian. The delta-function reversion proof explicitly records orientation and change-of-variables assumptions.

43.2 Proof-completeness policy

A displayed identity is treated in one of four ways: proved immediately; derived as a named corollary of an earlier proved theorem; marked explicitly as a historical or bibliographic fact; or supplied with an exact finite certificate algorithm. In particular, formulas newly imported from the supplemental directories are not left as unexplained tables. Their proofs use one of the reusable mechanisms established in the opening chapters: coefficient extraction, finite differences, labelled decomposition, Möbius inversion, repeated integration by parts, formal residues, or Euler–Maclaurin summation.

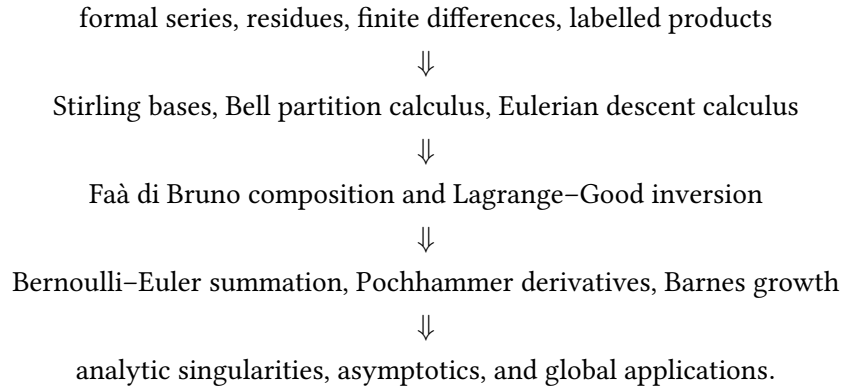
43.3 Implementation-neutral reproducibility

Every triangular array in the article can be generated from its recurrence in $O(N^2)$ arithmetic operations through row N . Bell-polynomial composition uses finite convolution; reversion uses either Lagrange coefficient extraction or triangular composition; Bernoulli, Euler, and Genocchi rows follow directly from their Appell generating functions; complementary Bell numbers follow from (30.4). These algorithms depend only on exact ring operations and are reproducible in any computer algebra system.

The LaTeX source and the rendered PDF distributed with this article are the canonical build artifacts. The PDF was compiled from the included source, not from one of the five pre-existing draft PDFs.

43.4 Final dependency map

The enlarged article is organized by the following chain:



This map is also a proof map: nearly every named formula in the five drafts and two supplemental directories is a specialization of one arrow in the chain.

Selected bibliography and provenance

Bibliography

- [1] E. T. Bell, “Exponential polynomials,” *Annals of Mathematics*, second series, vol. 35 (1934), pp. 258–277.
- [2] A. Z. Broder, “The r -Stirling numbers,” *Discrete Mathematics*, vol. 49 (1984), pp. 241–259.
- [3] C. A. Charalambides, *Enumerative Combinatorics*, Chapman & Hall/CRC, 2002.
- [4] A. Claesson, “Generalized pattern avoidance,” *European Journal of Combinatorics*, vol. 22 (2001), pp. 961–971.
- [5] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, translated by J. W. Nienhuys, D. Reidel, 1974.
- [6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [7] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, second edition, Addison–Wesley, 1994.
- [8] L. H. Harper, “Stirling behavior is asymptotically normal,” *Annals of Mathematical Statistics*, vol. 38 (1967), pp. 410–414.
- [9] P. Henrici, *Applied and Computational Complex Analysis*, vol. 1, Wiley, 1974.
- [10] L. C. Hsu and P. J.-S. Shiue, “A unified approach to generalized Stirling numbers,” *Advances in Applied Mathematics*, vol. 20 (1998), pp. 366–384.
- [11] W. P. Johnson, “The curious history of Faà di Bruno’s formula,” *American Mathematical Monthly*, vol. 109 (2002), pp. 217–234.
- [12] L. Moser and M. Wyman, “An asymptotic formula for the Bell numbers,” *Transactions of the Royal Society of Canada*, section III, vol. 49 (1955), pp. 49–54.
- [13] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, editors, *NIST Digital Library of Mathematical Functions*, National Institute of Standards and Technology, release 1.2.7, 15 June 2026, accessed 1 September 2026.
- [14] E. W. Barnes, “The theory of the G -function,” *Quarterly Journal of Pure and Applied Mathematics*, vol. 31 (1900), pp. 264–314.
- [15] I. M. Gessel, “Lagrange inversion,” *Journal of Combinatorial Theory, Series A*, vol. 144 (2016), pp. 212–249.
- [16] I. J. Good, “Generalizations to several variables of Lagrange’s expansion, with applications to stochastic processes,” *Proceedings of the Cambridge Philosophical Society*, vol. 56 (1960), pp. 367–380.

- [17] H.-K. Hwang, C.-Y. Li, and V. Zacharovas, “Elementary asymptotics for the Stirling numbers of the second kind: The central range,” preprint, arXiv:2605.29633, 2026.
- [18] N. E. Nörlund, *Vorlesungen über Differenzenrechnung*, Springer, 1924.
- [19] J. Riordan, *An Introduction to Combinatorial Analysis*, Wiley, 1958.
- [20] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
- [21] M. Z. Spivey, “A generalized recurrence for Bell numbers,” *Journal of Integer Sequences*, vol. 11 (2008), Article 08.2.5.
- [22] R. P. Stanley, *Enumerative Combinatorics*, vol. 1, second edition, Cambridge University Press, 2012.
- [23] N. M. Temme, “Asymptotic estimates of Stirling numbers,” *Studies in Applied Mathematics*, vol. 89 (1993), pp. 233–243.
- [24] H. S. Wilf, *generatingfunctionology*, third edition, A K Peters, 2005.
- [25] *Formulae-2.zip*, the source archive supplied for this project: five substantial LaTeX drafts in `articles/`, encyclopedia captures and text transcriptions in `Wikipedia/`, and MathWorld captures with Wolfram notebooks in `MathWorld/`.

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